



US 20250285351A1

(19) **United States**

(12) **Patent Application Publication**
Zohar et al.

(10) **Pub. No.: US 2025/0285351 A1**

(43) **Pub. Date: Sep. 11, 2025**

(54) **AUGMENTED REALITY TRY-ON
EXPERIENCE FOR FRIEND**

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(21) Appl. No.: **19/214,506**

(22) Filed: **May 21, 2025**

Related U.S. Application Data

(63) Continuation of application No. 18/104,840, filed on
Feb. 2, 2023, now Pat. No. 12,340,453.

Publication Classification

(51) **Int. Cl.**
G06T 13/40 (2011.01)
G06N 3/0475 (2023.01)
G06T 19/00 (2011.01)
G06T 19/20 (2011.01)

(52) **U.S. Cl.**

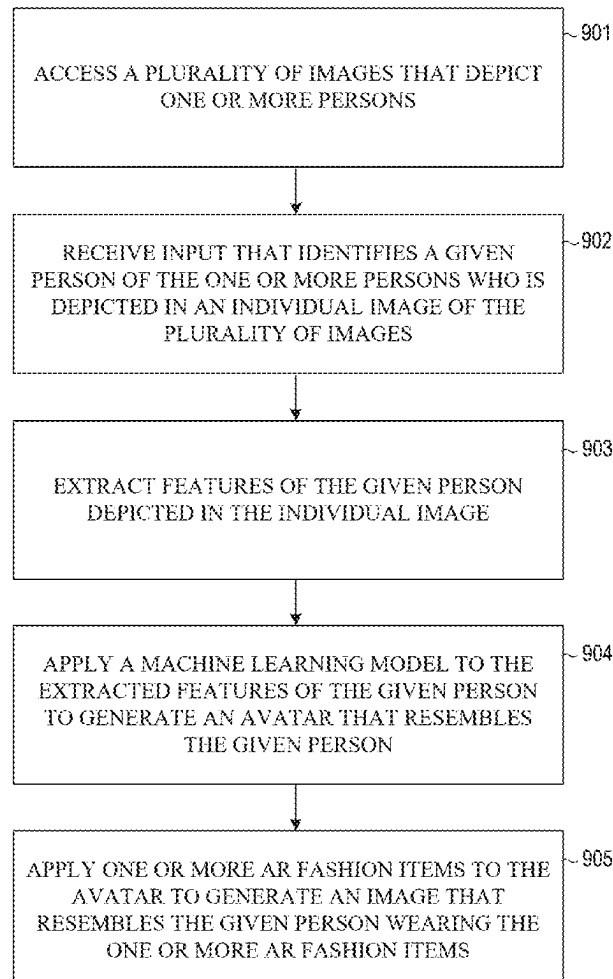
CPC **G06T 13/40** (2013.01); **G06N 3/0475**
(2023.01); **G06T 19/006** (2013.01); **G06T**
19/20 (2013.01); **G06T 2219/2012** (2013.01);
G06T 2219/2021 (2013.01)

(57)

ABSTRACT

Aspects of the present disclosure involve a system for providing an AR try-on experience for a friend. The system accesses a plurality of images that depict one or more persons. The system receives input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images. The system extracts features of the given person depicted in the individual image. The system applies a machine learning model to the extracted features of the given person to generate an avatar that resembles the given person. The system applies one or more augmented reality (AR) fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items.

900



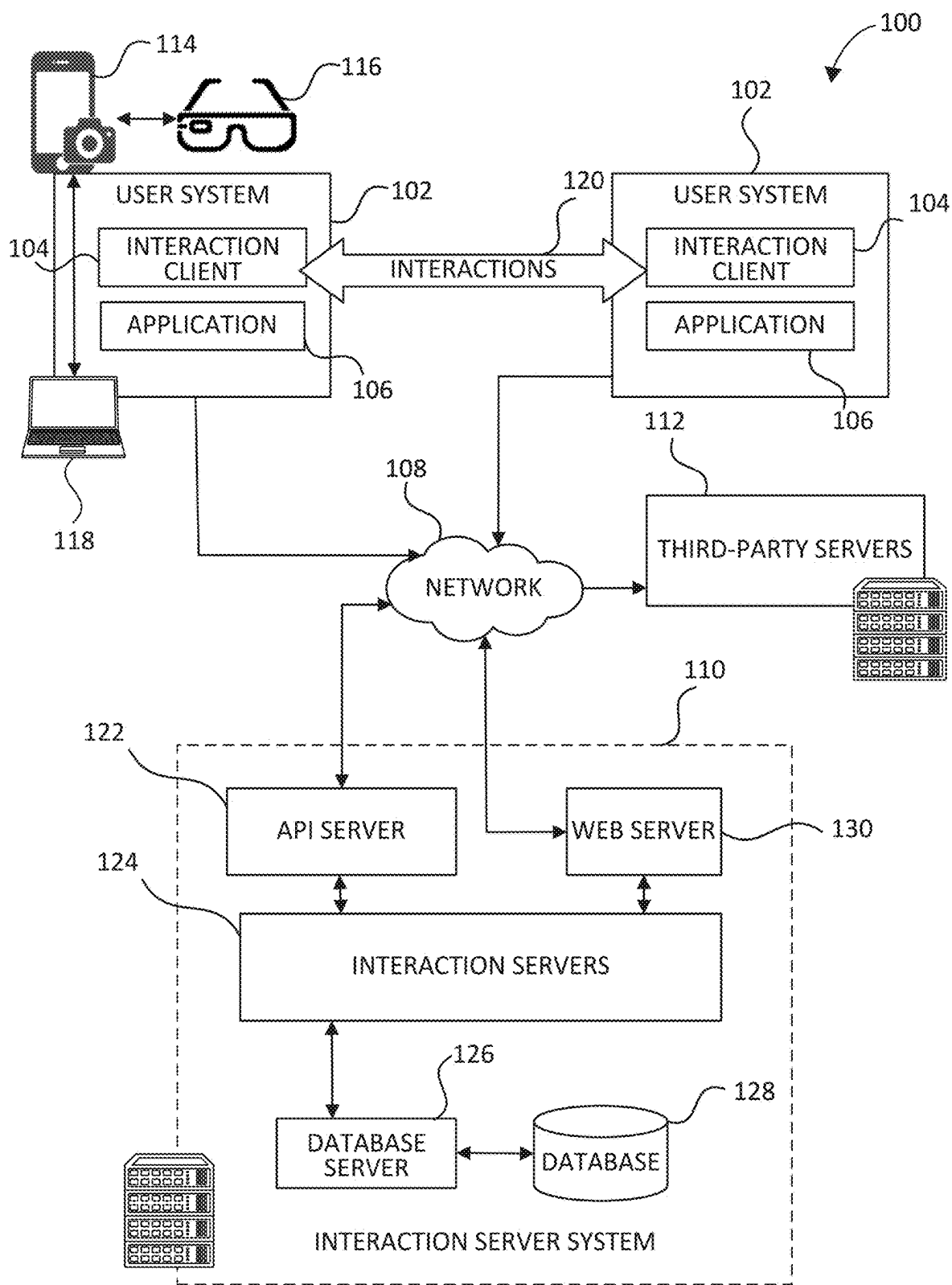


FIG. 1

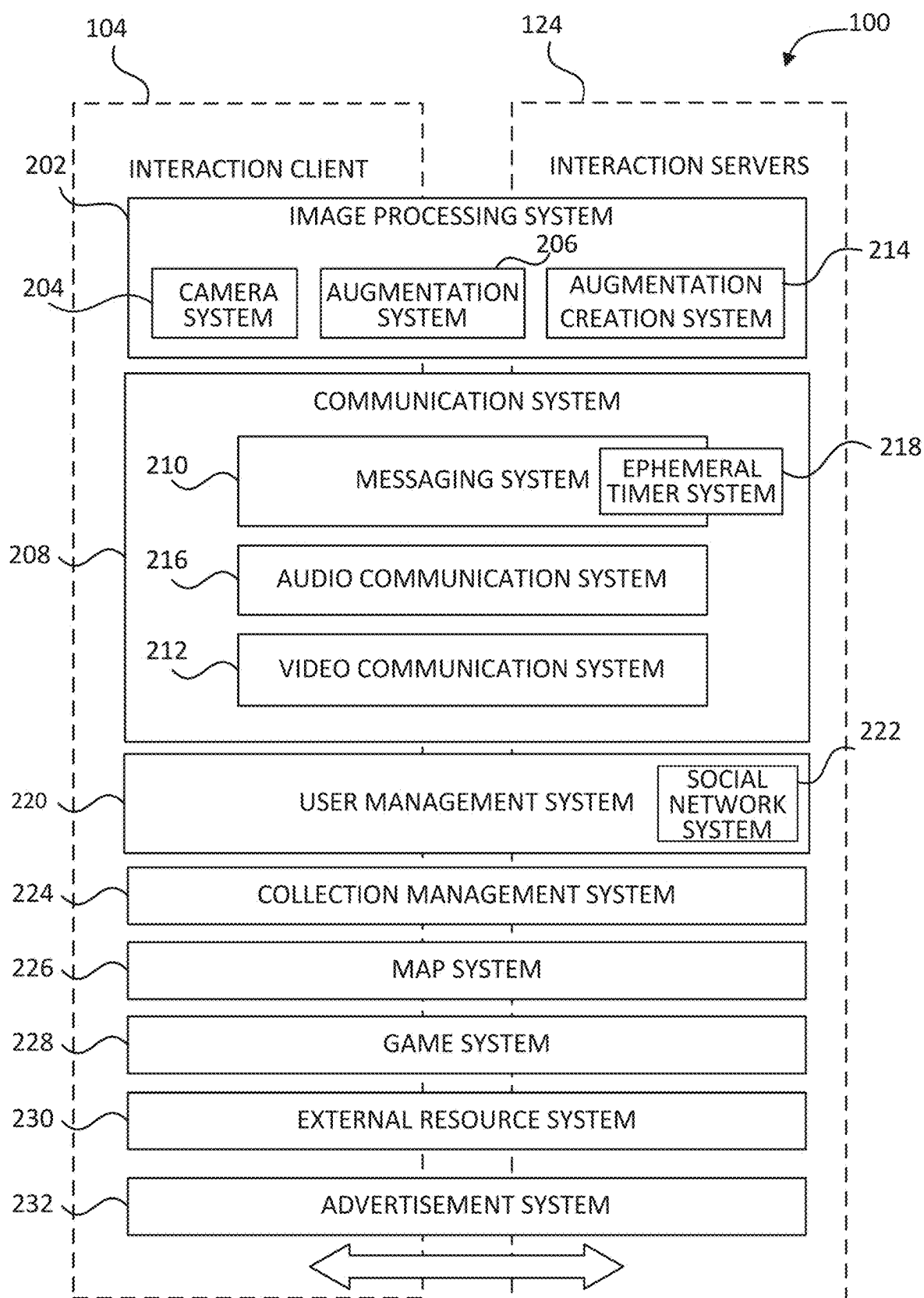


FIG. 2

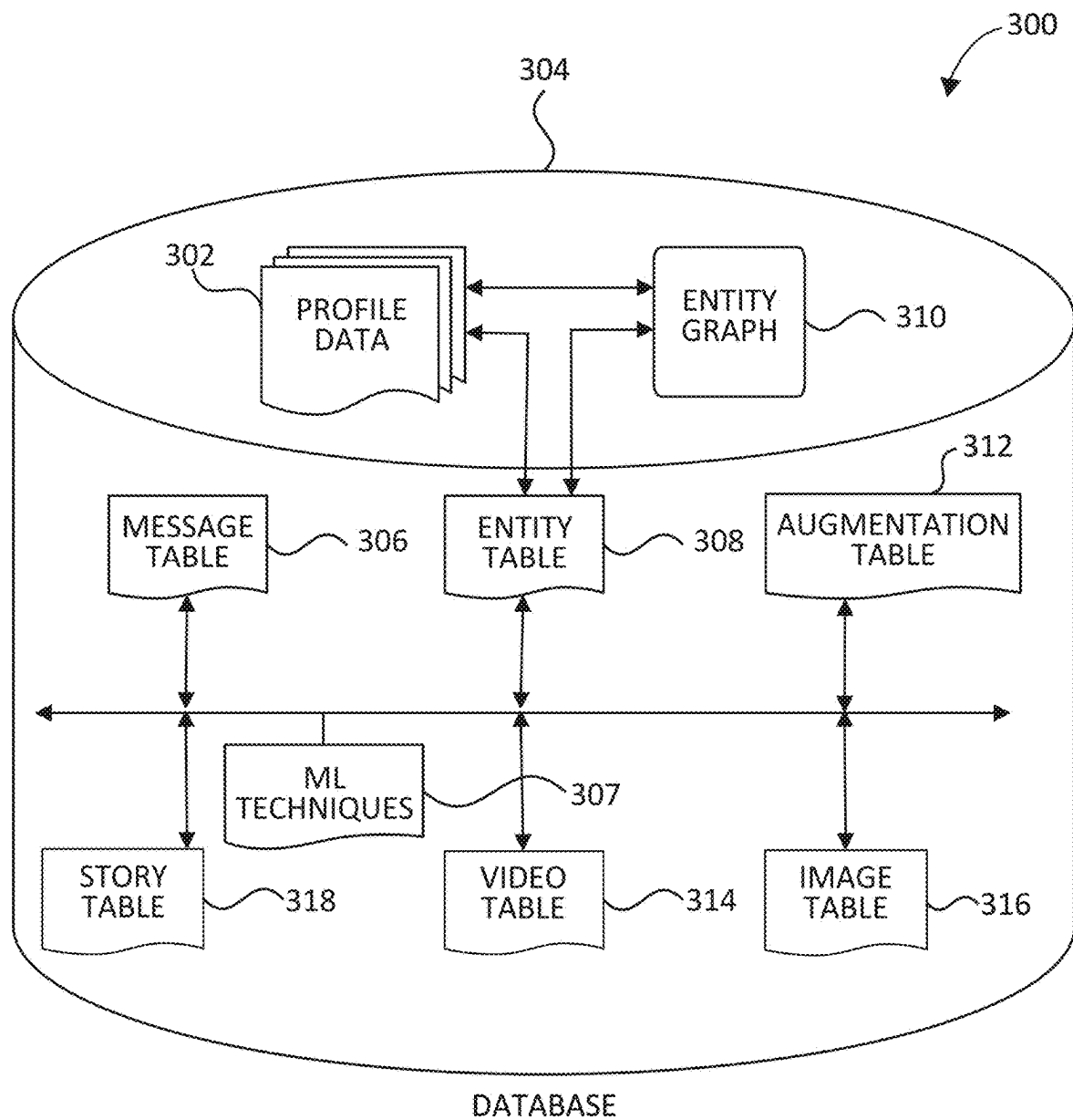


FIG. 3

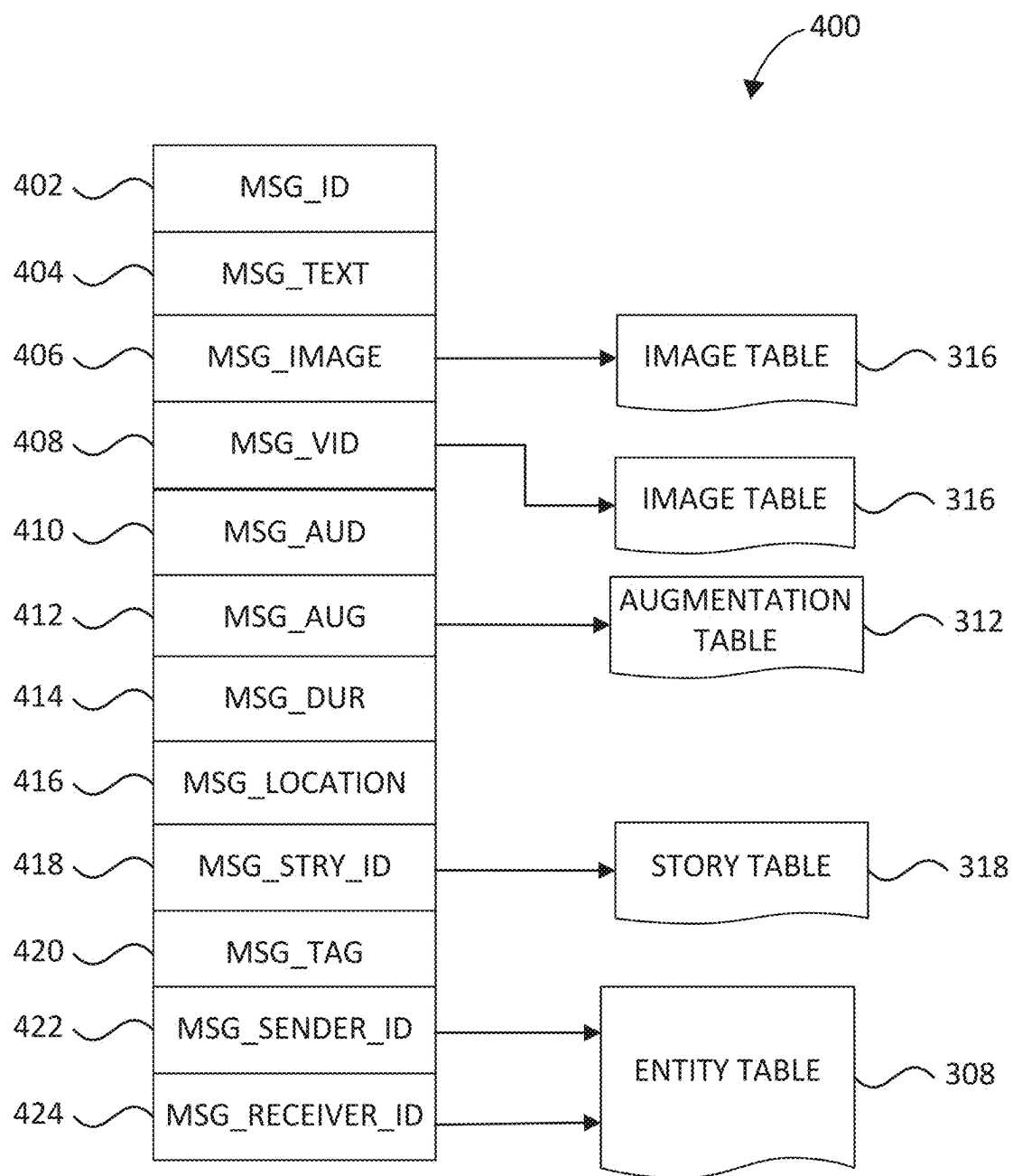


FIG. 4

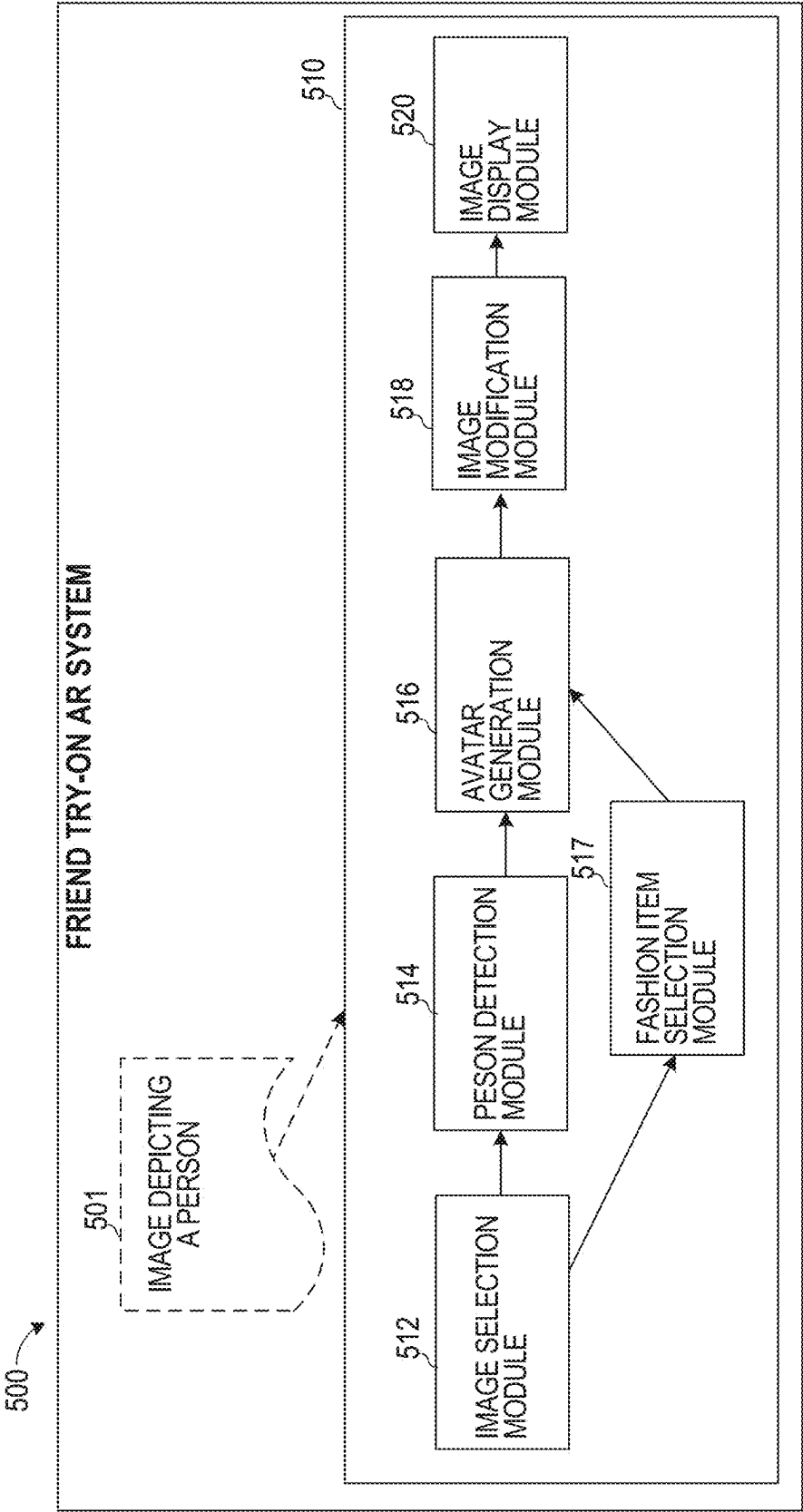


FIG. 5

600 →

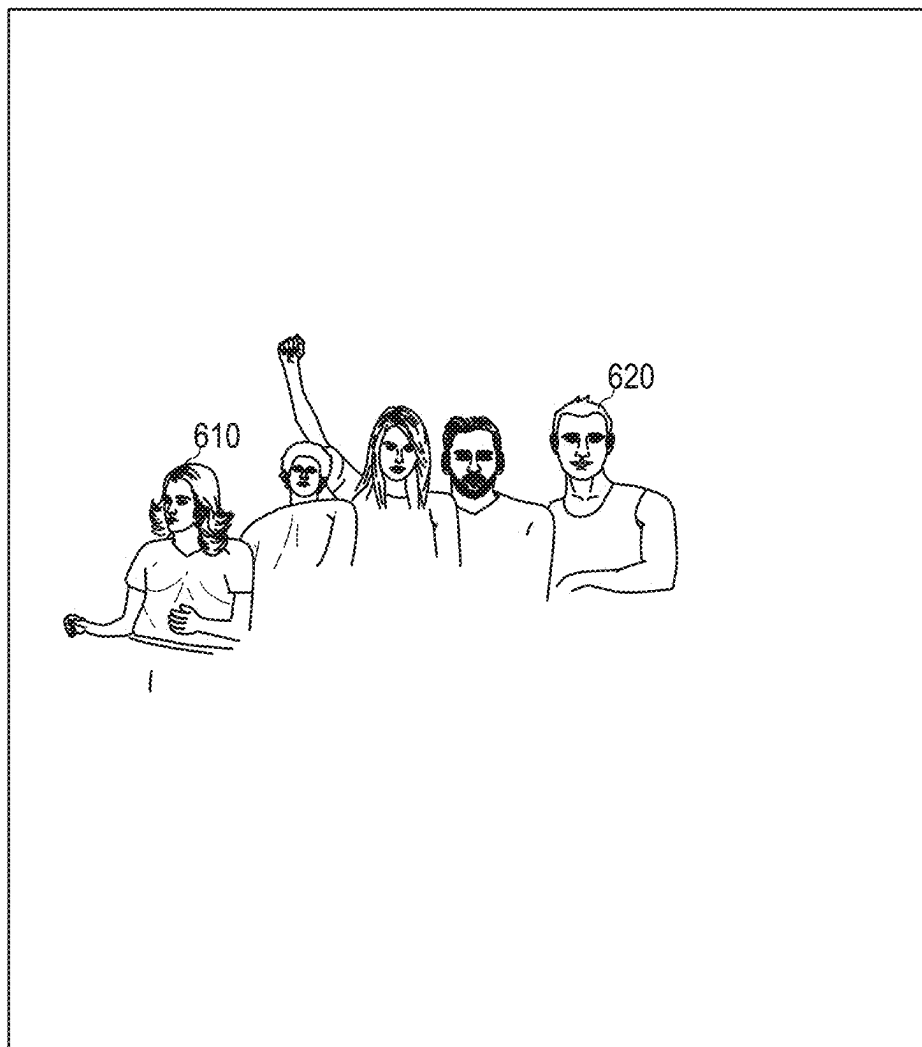


FIG. 6

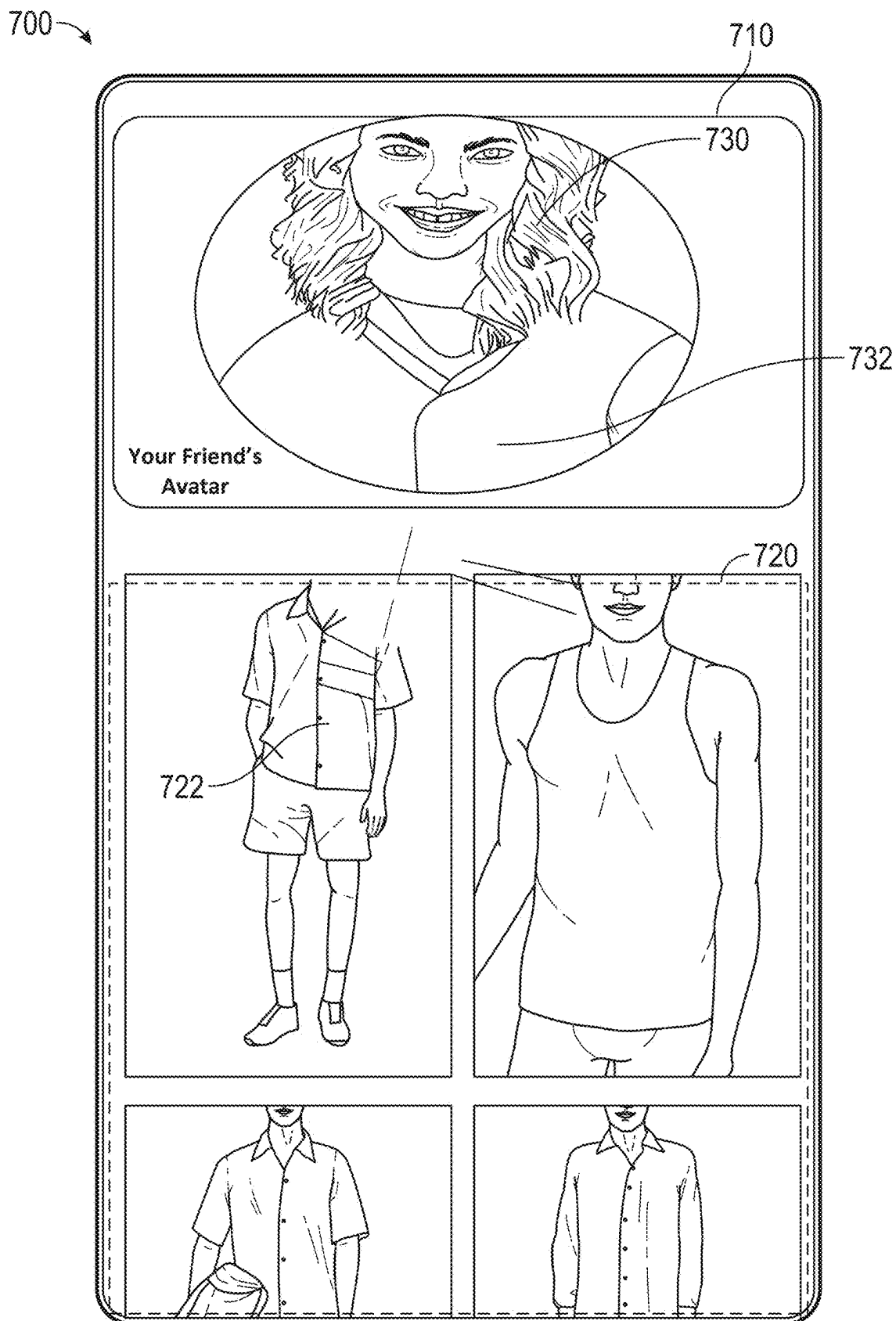


FIG. 7

800

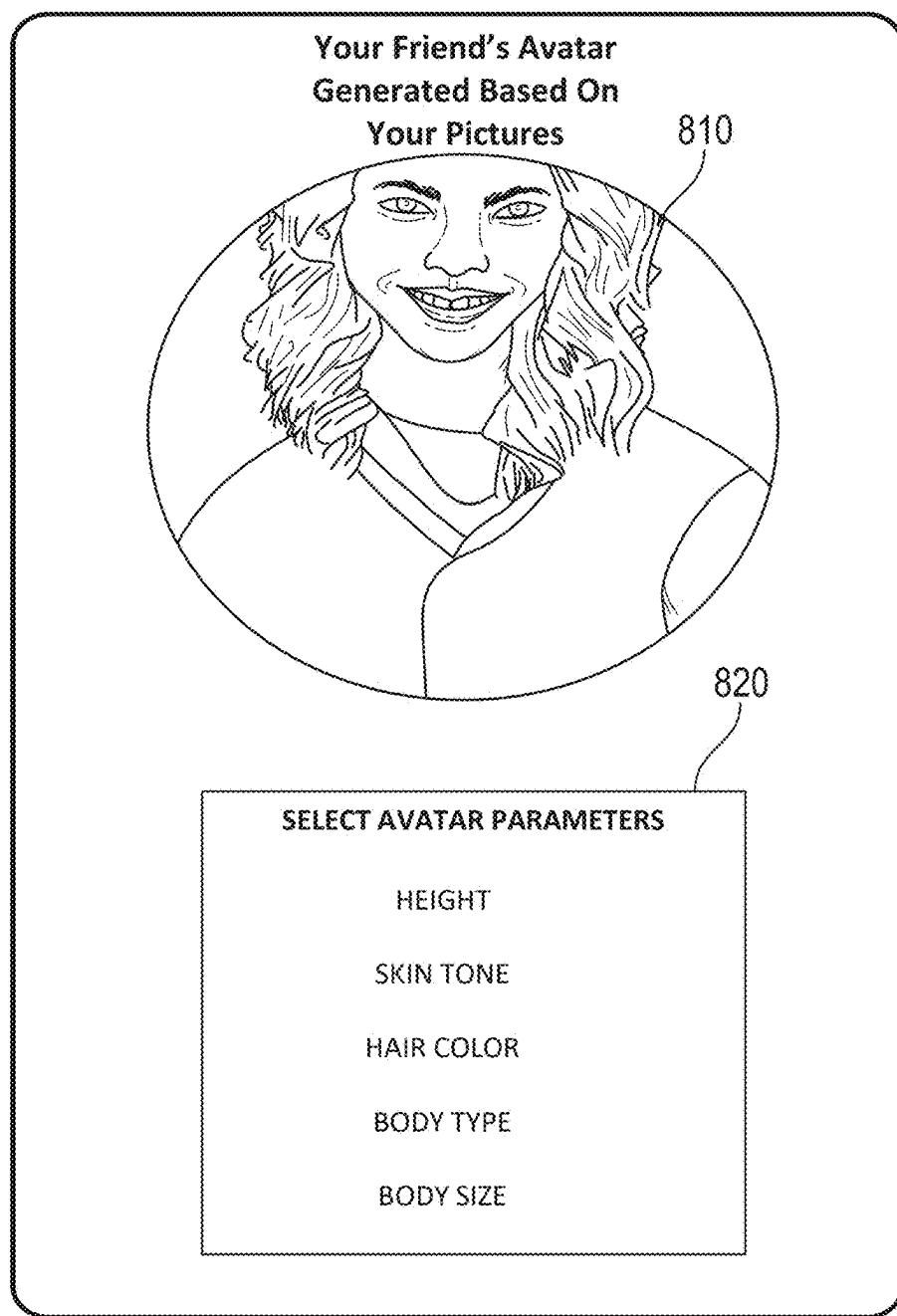


FIG. 8

900

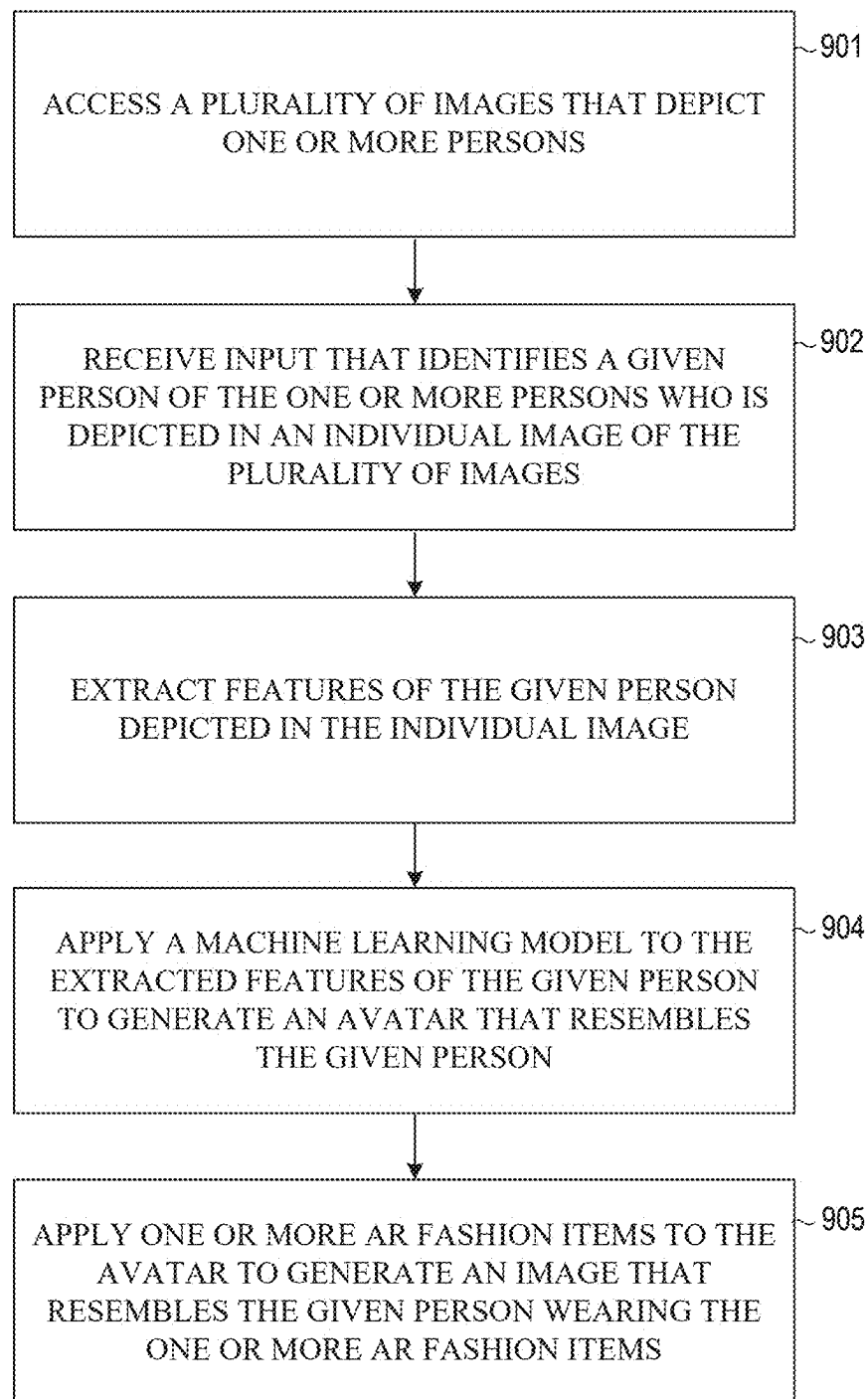


FIG. 9

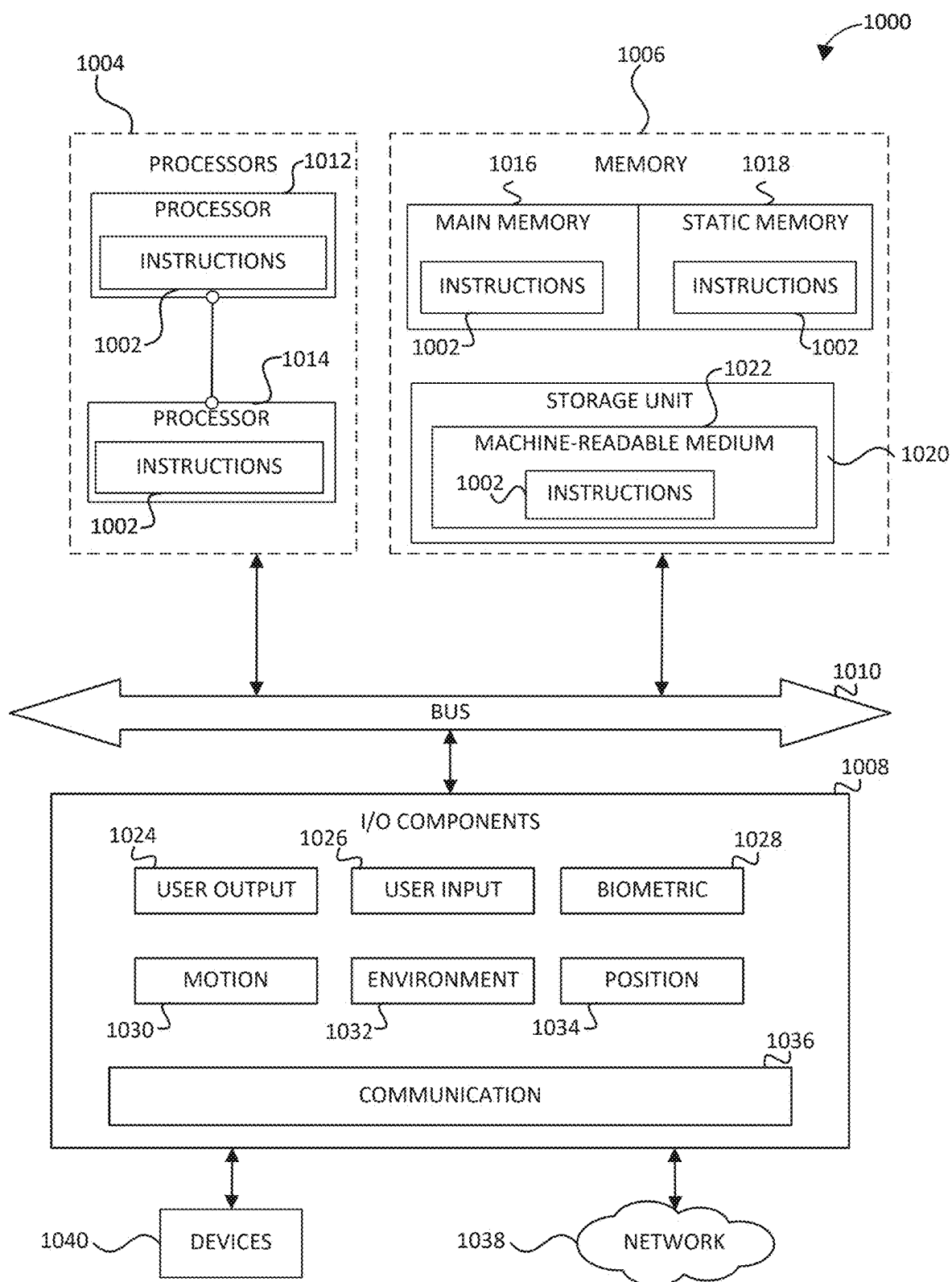


FIG. 10

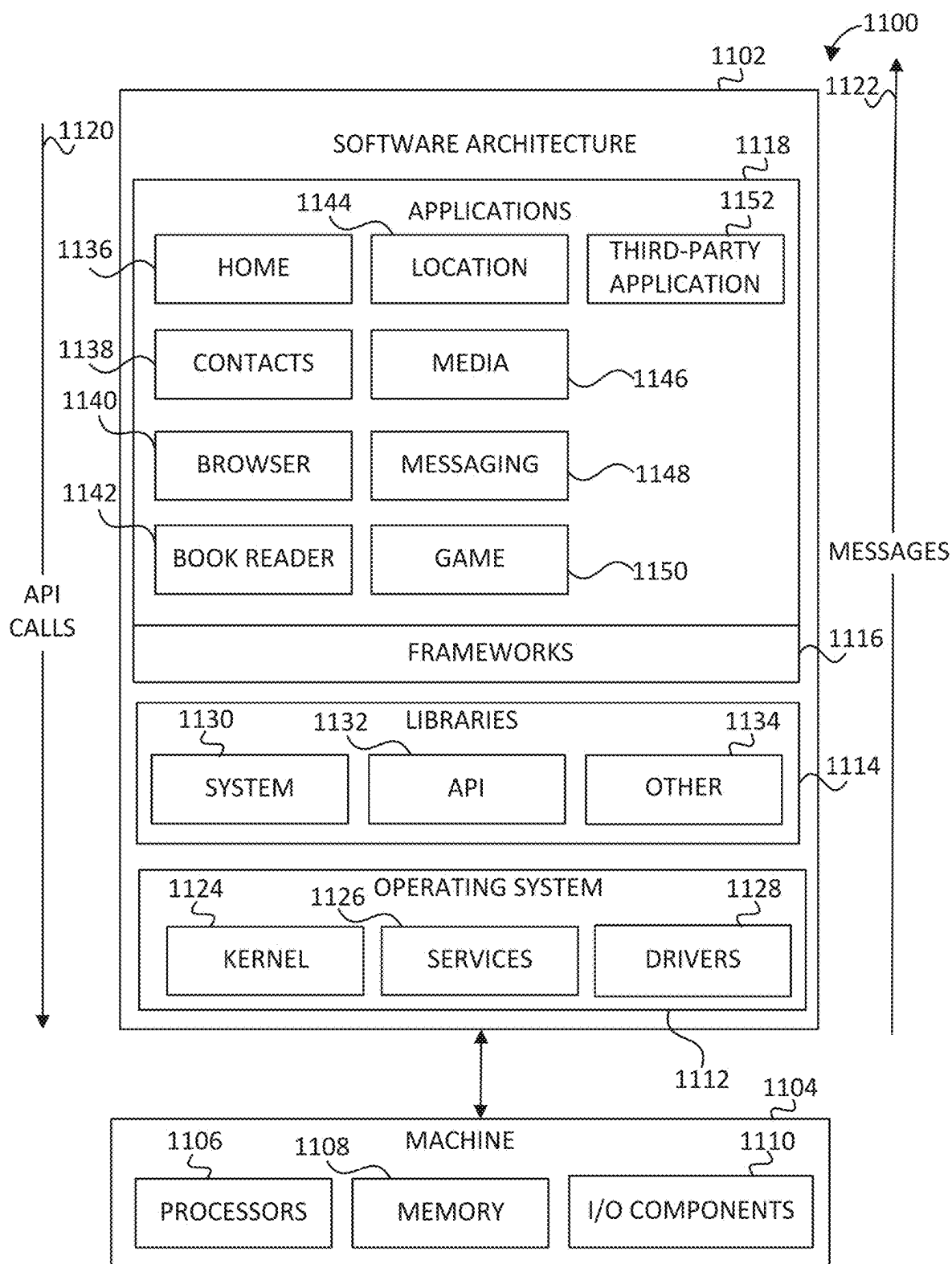


FIG. 11

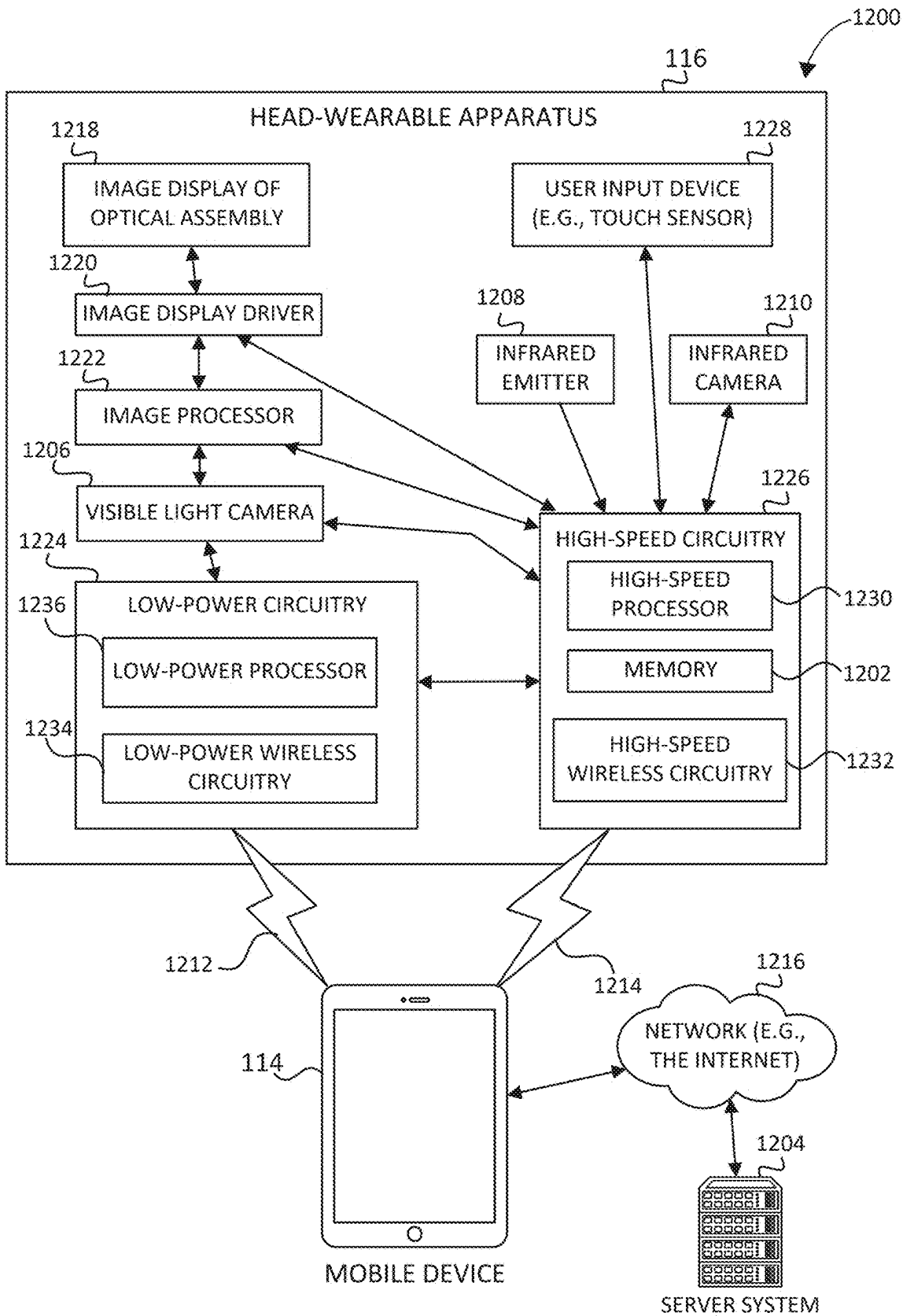


FIG. 12

AUGMENTED REALITY TRY-ON EXPERIENCE FOR FRIEND

CLAIM OF PRIORITY

[0001] This application is a continuation of U.S. patent application Ser. No. 18/104,840, filed on Feb. 2, 2023, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] The present disclosure relates generally to generating images and providing try-on experiences using an interaction application.

BACKGROUND

[0003] Augmented reality (AR) is a modification of a real-world environment with the addition or overlay of virtual content. For example, in virtual reality (VR), a user is completely immersed in a virtual world, whereas in AR, the user is immersed in a world where virtual objects are combined or superimposed on the real world. An AR system aims to generate and present virtual objects that interact realistically with a real-world environment and with each other. Examples of AR applications can include single or multiple player video games, instant messaging systems, and the like.

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0004] In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. To easily identify the discussion of any particular element or act, the most significant digit or digits in a reference number refer to the figure number in which that element is first introduced. Some nonlimiting examples are illustrated in the figures of the accompanying drawings in which:

[0005] FIG. 1 is a diagrammatic representation of a networked environment in which the present disclosure may be deployed, in accordance with some examples.

[0006] FIG. 2 is a diagrammatic representation of an interaction application, in accordance with some examples.

[0007] FIG. 3 is a diagrammatic representation of a data structure as maintained in a database, in accordance with some examples.

[0008] FIG. 4 is a diagrammatic representation of a message, in accordance with some examples.

[0009] FIG. 5 is a block diagram showing an example friend try-on AR system, according to some examples.

[0010] FIGS. 6-8 are diagrammatic representations of inputs and outputs of the friend try-on AR system, in accordance with some examples.

[0011] FIG. 9 is a flowchart illustrating example operations of the friend try-on AR system, according to some examples.

[0012] FIG. 10 is a diagrammatic representation of a machine in the form of a computer system within which a set of instructions may be executed to cause the machine to perform any one or more of the methodologies discussed herein, according to some examples.

[0013] FIG. 11 is a block diagram showing a software architecture within which examples may be implemented.

[0014] FIG. 12 illustrates a system in which the head-wearable apparatus, according to some examples.

DETAILED DESCRIPTION

[0015] The description that follows includes systems, methods, techniques, instruction sequences, and computing machine program products that embody illustrative examples of the disclosure. In the following description, for the purposes of explanation, numerous specific details are set forth in order to provide an understanding of various examples. It will be evident, however, to those skilled in the art, that examples may be practiced without these specific details. In general, well-known instruction instances, protocols, structures, and techniques are not necessarily shown in detail.

[0016] Typically, messaging applications and other social network platforms allow users to view a live or real-time camera feed that depicts the users wearing different items. For example, a user can activate a virtual try-on experience in which one or more AR items are placed on the user depicted in the image in real-time. This provides a user with the ability to visualize how different products look on the user before the user purchases the products. To activate such experiences, users can browse products through one software application, such as a website. The user can select a given product, such as a fashion item, depicted on the website. Then, the user is presented with an option to launch an AR shopping experience application in which the user can virtually try-on the selected product. A video depicting the user trying on the product is presented to the user in a separate application associated with the AR shopping experience. Namely, the user is navigated away from the website to a whole other experience to provide the user with the virtual try-on experience. The need to navigate the user away from the website makes online shopping difficult because the user has to continuously navigate back and forth between the AR try-on experience and the display of the products on the website. Having to navigate through a multitude of pages of information just to try on virtually certain products and to select new products to try on in a separate interface is discouraging and very disruptive. Also having to re-launch the AR experience each time a new product is selected consumes a great deal of resources and is incredibly inefficient.

[0017] In some cases, users may be interested in purchasing fashion items for other users or their friends. To do so, the users need to obtain access to their friend's avatar by explicitly requesting the friend to provide their avatar to the user. However, transmitting the avatar from one client device to another is incredibly time consuming and wastes time and system resources. Also, users may be interested in secretly shopping for their friends and, in such circumstances, the users may not be willing to explicitly request the friend to provide the avatar. There is no way for users to visualize how different fashion items will look on their friends before purchasing the corresponding items. Usually, users will purchase the items and only later find out that the items do not fit properly or do not look the way the users thought the items would look when the friend physically tries on the items. This results in wasted time and effort and missed purchasing opportunities.

[0018] The disclosed techniques improve the efficiency of using an electronic device by providing a virtual try-on experience in which an avatar for a friend or a second user is generated based on images that are readily available on a given client device of a first user. Namely, the first user can navigate and browse through various images that may have

been previously exchanged in messages between the first user and the second user. The first user can select one or more images that depict the second user (e.g., the target user) for whom the first user is interested in purchasing one or more fashion items. After selecting the images, a machine learning model is applied to features of the second user depicted in the selected images to generate an avatar that resembles the second user. The avatar can have a similar body size and shape as the second user and can be interacted with to apply one or more AR fashion items. This allows the first user to visualize how certain fashion items will look on the second user by way of the generated avatar of the second user before purchasing the corresponding items or merchandise. In this way, the overall user experience and AR experience is improved and the amount of resources needed to accomplish a task is reduced.

[0019] This can reduce the overall time and expense incurred by users trying on different fashion items, such as shoes, shirts, earrings, watches or other fashion items. As used herein, “article of clothing,” “fashion item,” and “garment” are used interchangeably and should be understood to have the same meaning. Article of clothing, garment, or fashion item can include a shirt, skirt, dress, shoe, purse, furniture item, household item, eyewear, eyeglasses, AR logos, AR emblems, pants, shorts, jackets, t-shirts, blouses, glasses, jewelry, earrings, bunny ears, a hat, earmuffs, or any other suitable item or wearable object.

[0020] Specifically, the disclosed techniques access a plurality of images that depict one or more persons. The disclosed techniques receive input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images. The disclosed techniques extract one or more features of the given person depicted in the individual image. The disclosed techniques apply a machine learning model to the extracted features of the given person to generate an avatar that resembles the given person. The disclosed techniques apply one or more augmented reality (AR) fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items. This improves the overall experience of the user in using the electronic device and reduces the overall amount of system resources needed to accomplish a task, such as by avoiding having to transmit an avatar over a network from one device to another.

Networked Computing Environment

[0021] FIG. 1 is a block diagram showing an example interaction system 100 for facilitating interactions (e.g., exchanging text messages, conducting text audio and video calls, or playing games) over a network. The interaction system 100 includes multiple user systems 102, each of which hosts multiple applications, including an interaction client 104 and other applications 106. Each interaction client 104 is communicatively coupled, via one or more communication networks including a network 108 (e.g., the Internet), to other instances of the interaction client 104 (e.g., hosted on respective other user systems 102), an interaction server system 110 and third-party servers 112. An interaction client 104 can also communicate with locally hosted applications 106 using Applications Program Interfaces (APIs).

[0022] Each user system 102 may include multiple user devices, such as a mobile device 114, head-wearable appa-

ratus 116, and a computer client device 118 that are communicatively connected to exchange data and messages.

[0023] An interaction client 104 interacts with other interaction clients 104 and with the interaction server system 110 via the network 108. The data exchanged between the interaction clients 104 (e.g., interactions 120) and between the interaction clients 104 and the interaction server system 110 includes functions (e.g., commands to invoke functions) and payload data (e.g., text, audio, video, or other multimedia data).

[0024] The interaction server system 110 provides server-side functionality via the network 108 to the interaction clients 104. While certain functions of the interaction system 100 are described herein as being performed by either an interaction client 104 or by the interaction server system 110, the location of certain functionality either within the interaction client 104 or the interaction server system 110 may be a design choice. For example, it may be technically preferable to initially deploy particular technology and functionality within the interaction server system 110 but to later migrate this technology and functionality to the interaction client 104 where a user system 102 has sufficient processing capacity.

[0025] The interaction server system 110 supports various services and operations that are provided to the interaction clients 104. Such operations include transmitting data to, receiving data from, and processing data generated by the interaction clients 104. This data may include message content, client device information, geolocation information, media augmentation and overlays, message content persistence conditions, social network information, and live event information. Data exchanges within the interaction system 100 are invoked and controlled through functions available via user interfaces (UIs) of the interaction clients 104.

[0026] Turning now specifically to the interaction server system 110, an Application Program Interface (API) server 122 is coupled to and provides programmatic interfaces to interaction servers 124, making the functions of the interaction servers 124 accessible to interaction clients 104, other applications 106 and third-party server 112. The interaction servers 124 are communicatively coupled to a database server 126, facilitating access to a database 128 that stores data associated with interactions processed by the interaction servers 124. Similarly, a web server 130 is coupled to the interaction servers 124 and provides web-based interfaces to the interaction servers 124. To this end, the web server 130 processes incoming network requests over the Hypertext Transfer Protocol (HTTP) and several other related protocols.

[0027] The Application Program Interface (API) server 122 receives and transmits interaction data (e.g., commands and message payloads) between the interaction servers 124 and the client systems 102 (and, for example, interaction clients 104 and other application 106) and the third-party server 112. Specifically, the Application Program Interface (API) server 122 provides a set of interfaces (e.g., routines and protocols) that can be called or queried by the interaction client 104 and other applications 106 to invoke functionality of the interaction servers 124. The Application Program Interface (API) server 122 exposes various functions supported by the interaction servers 124, including account registration; login functionality; the sending of interaction data, via the interaction servers 124, from a particular interaction client 104 to another interaction client

104; the communication of media files (e.g., images or video) from an interaction client **104** to the interaction servers **124**; the settings of a collection of media data (e.g., a story); the retrieval of a list of friends of a user of a user system **102**; the retrieval of messages and content; the addition and deletion of entities (e.g., friends) to an entity graph (e.g., a social graph); the location of friends within a social graph; and opening an application event (e.g., relating to the interaction client **104**).

[0028] The interaction servers **124** host multiple systems and subsystems, described below with reference to FIG. 2.

Linked Applications

[0029] Returning to the interaction client **104**, features and functions of an external resource (e.g., a linked application **106** or applet) are made available to a user via an interface of the interaction client **104**. In this context, “external” refers to the fact that the application **106** or applet is external to the interaction client **104**. The external resource is often provided by a third party but may also be provided by the creator or provider of the interaction client **104**. The interaction client **104** receives a user selection of an option to launch or access features of such an external resource. The external resource may be the application **106** installed on the user system **102** (e.g., a “native app”), or a small-scale version of the application (e.g., an “applet”) that is hosted on the user system **102** or remote of the user system **102** (e.g., on third-party servers **112**). The small-scale version of the application includes a subset of features and functions of the application (e.g., the full-scale, native version of the application) and is implemented using a markup-language document. In some examples, the small-scale version of the application (e.g., an “applet”) is a web-based, markup-language version of the application and is embedded in the interaction client **104**. In addition to using markup-language documents (e.g., a *.ml file), an applet may incorporate a scripting language (e.g., a *.js file or a *.json file) and a style sheet (e.g., a *.ss file).

[0030] In response to receiving a user selection of the option to launch or access features of the external resource, the interaction client **104** determines whether the selected external resource is a web-based external resource or a locally-installed application **106**. In some cases, applications **106** that are locally installed on the user system **102** can be launched independently of and separately from the interaction client **104**, such as by selecting an icon corresponding to the application **106** on a home screen of the user system **102**. Small-scale versions of such applications can be launched or accessed via the interaction client **104** and, in some examples, no or limited portions of the small-scale application can be accessed outside of the interaction client **104**. The small-scale application can be launched by the interaction client **104** receiving, from a third-party server **112** for example, a markup-language document associated with the small-scale application and processing such a document.

[0031] In response to determining that the external resource is a locally-installed application **106**, the interaction client **104** instructs the user system **102** to launch the external resource by executing locally-stored code corresponding to the external resource. In response to determining that the external resource is a web-based resource, the interaction client **104** communicates with the third-party servers **112** (for example) to obtain a markup-language

document corresponding to the selected external resource. The interaction client **104** then processes the obtained markup-language document to present the web-based external resource within a user interface of the interaction client **104**.

[0032] The interaction client **104** can notify a user of the user system **102**, or other users related to such a user (e.g., “friends”), of activity taking place in one or more external resources. For example, the interaction client **104** can provide participants in a conversation (e.g., a chat session) in the interaction client **104** with notifications relating to the current or recent use of an external resource by one or more members of a group of users. One or more users can be invited to join in an active external resource or to launch a recently-used but currently inactive (in the group of friends) external resource. The external resource can provide participants in a conversation, each using respective interaction clients **104**, with the ability to share an item, status, state, or location in an external resource in a chat session with one or more members of a group of users. The shared item may be an interactive chat card with which members of the chat can interact, for example, to launch the corresponding external resource, view specific information within the external resource, or take the member of the chat to a specific location or state within the external resource. Within a given external resource, response messages can be sent to users on the interaction client **104**. The external resource can selectively include different media items in the responses, based on a current context of the external resource.

[0033] The interaction client **104** can present a list of the available external resources (e.g., applications **106** or applets) to a user to launch or access a given external resource. This list can be presented in a context-sensitive menu. For example, the icons representing different ones of the application **106** (or applets) can vary based on how the menu is launched by the user (e.g., from a conversation interface or from a non-conversation interface).

System Architecture

[0034] FIG. 2 is a block diagram illustrating further details regarding the interaction system **100**, according to some examples. Specifically, the interaction system **100** is shown to comprise the interaction client **104** and the interaction servers **124**. The interaction system **100** embodies multiple subsystems, which are supported on the client-side by the interaction client **104** and on the server-side by the interaction servers **124**. Example subsystems are discussed below and can include a friend try-on AR system **500** that enables a user to launch an AR experience to overlay one or more AR fashion items on a friend’s avatar. An illustrative implementation of the friend try-on AR system **500** is shown and described in connection with FIG. 5 below.

[0035] Specifically, the friend try-on AR system **500** is a component that can be accessed by an AR/VR application implemented on the user system **102**. The AR/VR application uses an RGB camera to capture a monocular image of a real-world object. The AR/VR application applies various trained machine learning techniques or machine learning models on the captured image of the real-world object to generate body landmark features representing the real-world object depicted in the images or videos and to apply one or more AR visual effects to the captured image or video based on body landmark features. In some implementations, the AR/VR application continuously captures images of the user

and updates the body landmark features in real time or periodically to continuously or periodically update the applied one or more visual effects. This allows the user to move around in the real world and see the one or more visual effects update in real time.

[0036] An image processing system **202** provides various functions that enable a user to capture and augment (e.g., annotate or otherwise modify or edit) media content associated with a message.

[0037] A camera system **204** includes control software (e.g., in a camera application) that interacts with and controls hardware camera hardware (e.g., directly or via operating system controls) of the user system **102** to modify and augment real-time images captured and displayed via the interaction client **104**.

[0038] The augmentation system **206** provides functions related to the generation and publishing of augmentations (e.g., media overlays) for images captured in real-time by cameras of the user system **102** or retrieved from memory of the user system **102**. For example, the augmentation system **206** operatively selects, presents, and displays media overlays (e.g., an image filter or an image lens) to the interaction client **104** for the augmentation of real-time images received via the camera system **204** or stored images retrieved from memory **1006** of a user system **102**. These augmentations are selected by the augmentation system **206** and presented to a user of an interaction client **104**, based on a number of inputs and data, such as for example:

[0039] Geolocation of the user system **102**; and

[0040] Social network information of the user of the user system **102**.

[0041] An augmentation may include audio and visual content and visual effects. Examples of audio and visual content include pictures, texts, logos, animations, and sound effects. An example of a visual effect includes color overlaying. The audio and visual content or the visual effects can be applied to a media content item (e.g., a photo or video) at user system **102** for communication in a message, or applied to video content, such as a video content stream or feed transmitted from an interaction client **104**. As such, the image processing system **202** may interact with, and support, the various subsystems of the communication system **208**, such as the messaging system **210** and the video communication system **212**.

[0042] A media overlay may include text or image data that can be overlaid on top of a photograph taken by the user system **102** or a video stream produced by the user system **102**. In some examples, the media overlay may be a location overlay (e.g., Venice beach), a name of a live event, or a name of a merchant overlay (e.g., Beach Coffee House). In further examples, the image processing system **202** uses the geolocation of the user system **102** to identify a media overlay that includes the name of a merchant at the geolocation of the user system **102**. The media overlay may include other indicia associated with the merchant. The media overlays may be stored in the databases **128** and accessed through the database server **126**.

[0043] The image processing system **202** provides a user-based publication platform that enables users to select a geolocation on a map and upload content associated with the selected geolocation. The user may also specify circumstances under which a particular media overlay should be offered to other users. The image processing system **202**

generates a media overlay that includes the uploaded content and associates the uploaded content with the selected geolocation.

[0044] The augmentation creation system **214** supports augmented reality developer platforms and includes an application for content creators (e.g., artists and developers) to create and publish augmentations (e.g., augmented reality experiences) of the interaction client **104**. The augmentation creation system **214** provides a library of built-in features and tools to content creators including, for example custom shaders, tracking technology, and templates.

[0045] In some examples, the augmentation creation system **214** provides a merchant-based publication platform that enables merchants to select a particular augmentation associated with a geolocation via a bidding process. For example, the augmentation creation system **214** associates a media overlay of the highest bidding merchant with a corresponding geolocation for a predefined amount of time.

[0046] A communication system **208** is responsible for enabling and processing multiple forms of communication and interaction within the interaction system **100** and includes a messaging system **210**, an audio communication system **216**, and a video communication system **212**. The messaging system **210** is responsible for enforcing the temporary or time-limited access to content by the interaction clients **104**. The messaging system **210** incorporates multiple timers (e.g., within an ephemeral timer system **218**) that, based on duration and display parameters associated with a message or collection of messages (e.g., a story), selectively enable access (e.g., for presentation and display) to messages and associated content via the interaction client **104**. Further details regarding the operation of the ephemeral timer system **218** are provided below. The audio communication system **216** enables and supports audio communications (e.g., real-time audio chat) between multiple interaction clients **104**. Similarly, the video communication system **212** enables and supports video communications (e.g., real-time video chat) between multiple interaction clients **104**.

[0047] A user management system **220** is operationally responsible for the management of user data and profiles, and includes a social network system **222** that maintains information regarding relationships between users of the interaction system **100**.

[0048] A collection management system **224** is operationally responsible for managing sets or collections of media (e.g., collections of text, image video, and audio data). A collection of content (e.g., messages, including images, video, text, and audio) may be organized into an “event gallery” or an “event story.” Such a collection may be made available for a specified time period, such as the duration of an event to which the content relates. For example, content relating to a music concert may be made available as a “story” for the duration of that music concert. The collection management system **224** may also be responsible for publishing an icon that provides notification of a particular collection to the user interface of the interaction client **104**. The collection management system **224** includes a curation function that allows a collection manager to manage and curate a particular collection of content. For example, the curation interface enables an event organizer to curate a collection of content relating to a specific event (e.g., to delete inappropriate content or redundant messages). Additionally, the collection management system **224** employs machine vision (or image recognition technology) and con-

tent rules to curate a content collection automatically. In certain examples, compensation may be paid to a user to include user-generated content into a collection. In such cases, the collection management system 224 operates to automatically make payments to such users to use their content.

[0049] A map system 226 provides various geographic location functions and supports the presentation of map-based media content and messages by the interaction client 104. For example, the map system 226 enables the display of user icons or avatars (e.g., stored in profile data 302) on a map to indicate a current or past location of “friends” of a user, as well as media content (e.g., collections of messages including photographs and videos) generated by such friends, within the context of a map. For example, a message posted by a user to the interaction system 100 from a specific geographic location may be displayed within the context of a map at that particular location to “friends” of a specific user on a map interface of the interaction client 104. A user can furthermore share his or her location and status information (e.g., using an appropriate status avatar) with other users of the interaction system 100 via the interaction client 104, with this location and status information being similarly displayed within the context of a map interface of the interaction client 104 to selected users.

[0050] A game system 228 provides various gaming functions within the context of the interaction client 104. The interaction client 104 provides a game interface providing a list of available games that can be launched by a user within the context of the interaction client 104 and played with other users of the interaction system 100. The interaction system 100 further enables a particular user to invite other users to participate in the play of a specific game by issuing invitations to such other users from the interaction client 104. The interaction client 104 also supports audio, video, and text messaging (e.g., chats) within the context of gameplay, provides a leaderboard for the games, and also supports the provision of in-game rewards (e.g., coins and items).

[0051] An external resource system 230 provides an interface for the interaction client 104 to communicate with remote servers (e.g., third-party servers 112) to launch or access external resources, i.e., applications or applets. Each third-party server 112 hosts, for example, a markup language (e.g., HTML5) based application or a small-scale version of an application (e.g., game, utility, payment, or ride-sharing application). The interaction client 104 may launch a web-based resource (e.g., application) by accessing the HTML5 file from the third-party servers 112 associated with the web-based resource. Applications hosted by third-party servers 112 are programmed in JavaScript leveraging a Software Development Kit (SDK) provided by the interaction servers 124. The SDK includes Application Programming Interfaces (APIs) with functions that can be called or invoked by the web-based application. The interaction servers 124 host a JavaScript library that provides a given external resource access to specific user data of the interaction client 104. HTML5 is an example of technology for programming games, but applications and resources programmed based on other technologies can be used.

[0052] To integrate the functions of the SDK into the web-based resource, the SDK is downloaded by the third-party server 112 from the interaction servers 124 or is otherwise received by the third-party server 112. Once downloaded or received, the SDK is included as part of the

application code of a web-based external resource. The code of the web-based resource can then call or invoke certain functions of the SDK to integrate features of the interaction client 104 into the web-based resource.

[0053] The SDK stored on the interaction server system 110 effectively provides the bridge between an external resource (e.g., applications 106 or applets) and the interaction client 104. This gives the user a seamless experience of communicating with other users on the interaction client 104 while also preserving the look and feel of the interaction client 104. To bridge communications between an external resource and an interaction client 104, the SDK facilitates communication between third-party servers 112 and the interaction client 104. A Web ViewJavaScriptBridge running on a user system 102 establishes two one-way communication channels between an external resource and the interaction client 104. Messages are sent between the external resource and the interaction client 104 via these communication channels asynchronously. Each SDK function invocation is sent as a message and callback. Each SDK function is implemented by constructing a unique callback identifier and sending a message with that callback identifier.

[0054] By using the SDK, not all information from the interaction client 104 is shared with third-party servers 112. The SDK limits which information is shared based on the needs of the external resource. Each third-party server 112 provides an HTML5 file corresponding to the web-based external resource to interaction servers 124. The interaction servers 124 can add a visual representation (such as a box art or other graphic) of the web-based external resource in the interaction client 104. Once the user selects the visual representation or instructs the interaction client 104 through a GUI of the interaction client 104 to access features of the web-based external resource, the interaction client 104 obtains the HTML5 file and instantiates the resources to access the features of the web-based external resource.

[0055] The interaction client 104 presents a graphical user interface (e.g., a landing page or title screen) for an external resource. During, before, or after presenting the landing page or title screen, the interaction client 104 determines whether the launched external resource has been previously authorized to access user data of the interaction client 104. In response to determining that the launched external resource has been previously authorized to access user data of the interaction client 104, the interaction client 104 presents another graphical user interface of the external resource that includes functions and features of the external resource. In response to determining that the launched external resource has not been previously authorized to access user data of the interaction client 104, after a threshold period of time (e.g., 3 seconds) of displaying the landing page or title screen of the external resource, the interaction client 104 slides up (e.g., animates a menu as surfacing from a bottom of the screen to a middle or other portion of the screen) a menu for authorizing the external resource to access the user data. The menu identifies the type of user data that the external resource will be authorized to use. In response to receiving a user selection of an accept option, the interaction client 104 adds the external resource to a list of authorized external resources and allows the external resource to access user data from the interaction client 104. The external resource is authorized by the interaction client 104 to access the user data under an OAuth 2 framework.

[0056] The interaction client **104** controls the type of user data that is shared with external resources based on the type of external resource being authorized. For example, external resources that include full-scale applications (e.g., an application **106**) are provided with access to a first type of user data (e.g., two-dimensional avatars of users with or without different avatar characteristics). As another example, external resources that include small-scale versions of applications (e.g., web-based versions of applications) are provided with access to a second type of user data (e.g., payment information, two-dimensional avatars of users, three-dimensional avatars of users, and avatars with various avatar characteristics). Avatar characteristics include different ways to customize a look and feel of an avatar, such as different poses, facial features, clothing, and so forth.

[0057] An advertisement system **232** operationally enables the purchasing of advertisements by third parties for presentation to end-users via the interaction clients **104** and also handles the delivery and presentation of these advertisements.

[0058] A friend try-on AR system **500** accesses a plurality of images that depict one or more persons. The friend try-on AR system **500** receives input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images. The friend try-on AR system **500** extracts features of the given person depicted in the individual image. The friend try-on AR system **500** applies a machine learning model to the extracted features of the given person to generate an avatar that resembles the given person. The friend try-on AR system **500** applies one or more AR fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items.

[0059] An illustrative implementation of the friend try-on AR system **500** is shown and described in connection with FIG. **5** below.

Data Architecture

[0060] FIG. **3** is a schematic diagram illustrating data structures **300**, which may be stored in the database **304** of the interaction server system **110**, according to certain examples. While the content of the database **304** is shown to comprise multiple tables, it will be appreciated that the data could be stored in other types of data structures (e.g., as an object-oriented database).

[0061] The database **304** includes message data stored within a message table **306**. This message data includes, for any particular message, at least message sender data, message recipient (or receiver) data, and a payload. Further details regarding information that may be included in a message, and included within the message data stored in the message table **306**, are described below with reference to FIG. **3**.

[0062] An entity table **308** stores entity data, and is linked (e.g., referentially) to an entity graph **310** and profile data **302**. Entities for which records are maintained within the entity table **308** may include individuals, corporate entities, organizations, objects, places, events, and so forth. Regardless of entity type, any entity regarding which the interaction server system **110** stores data may be a recognized entity. Each entity is provided with a unique identifier, as well as an entity type identifier (not shown).

[0063] The entity graph **310** stores information regarding relationships and associations between entities. Such rela-

tionships may be social, professional (e.g., work at a common corporation or organization), interest-based, or activity-based, merely for example. Certain relationships between entities may be unidirectional, such as a subscription by an individual user to digital content of a commercial or publishing user (e.g., a newspaper or other digital media outlet, or a brand). Other relationships may be bidirectional, such as a “friend” relationship between individual users of the interaction system **100**.

[0064] Certain permissions and relationships may be attached to each relationship, and also to each direction of a relationship. For example, a bidirectional relationship (e.g., a friend relationship between individual users) may include authorization for the publication of digital content items between the individual users, but may impose certain restrictions or filters on the publication of such digital content items (e.g., based on content characteristics, location data or time of day data). Similarly, a subscription relationship between an individual user and a commercial user may impose different degrees of restrictions on the publication of digital content from the commercial user to the individual user, and may significantly restrict or block the publication of digital content from the individual user to the commercial user. A particular user, as an example of an entity, may record certain restrictions (e.g., by way of privacy settings) in a record for that entity within the entity table **308**. Such privacy settings may be applied to all types of relationships within the context of the interaction system **100**, or may selectively be applied to certain types of relationships.

[0065] The profile data **302** stores multiple types of profile data about a particular entity. The profile data **302** may be selectively used and presented to other users of the interaction system **100** based on privacy settings specified by a particular entity. Where the entity is an individual, the profile data **302** includes, for example, a user name, telephone number, address, settings (e.g., notification and privacy settings), as well as a user-selected avatar representation (or collection of such avatar representations). A particular user may then selectively include one or more of these avatar representations within the content of messages communicated via the interaction system **100**, and on map interfaces displayed by interaction clients **104** to other users. The collection of avatar representations may include “status avatars,” which present a graphical representation of a status or activity that the user may select to communicate at a particular time.

[0066] Where the entity is a group, the profile data **302** for the group may similarly include one or more avatar representations associated with the group, in addition to the group name, members, and various settings (e.g., notifications) for the relevant group.

[0067] The database **304** also stores augmentation data, such as overlays or filters, in an augmentation table **312**. The augmentation data is associated with and applied to videos (for which data is stored in a video table **314**) and images (for which data is stored in an image table **316**).

[0068] Filters, in some examples, are overlays that are displayed as overlaid on an image or video during presentation to a recipient user. Filters may be of various types, including user-selected filters from a set of filters presented to a sending user by the interaction client **104** when the sending user is composing a message. Other types of filters include geolocation filters (also known as geo-filters), which may be presented to a sending user based on geographic

location. For example, geolocation filters specific to a neighborhood or special location may be presented within a user interface by the interaction client **104**, based on geolocation information determined by a Global Positioning System (GPS) unit of the user system **102**.

[0069] Another type of filter is a data filter, which may be selectively presented to a sending user by the interaction client **104** based on other inputs or information gathered by the user system **102** during the message creation process. Examples of data filters include current temperature at a specific location, a current speed at which a sending user is traveling, battery life for a user system **102**, or the current time.

[0070] Other augmentation data that may be stored within the image table **316** includes augmented reality content items (e.g., corresponding to applying “lenses” or augmented reality experiences). An augmented reality content item may be a real-time special effect and sound that may be added to an image or a video.

[0071] A story table **318** stores data regarding collections of messages and associated image, video, or audio data, which are compiled into a collection (e.g., a story or a gallery). The creation of a particular collection may be initiated by a particular user (e.g., each user for which a record is maintained in the entity table **308**). A user may create a “personal story” in the form of a collection of content that has been created and sent/broadcast by that user. To this end, the user interface of the interaction client **104** may include an icon that is user-selectable to enable a sending user to add specific content to his or her personal story.

[0072] A collection may also constitute a “live story,” which is a collection of content from multiple users that is created manually, automatically, or using a combination of manual and automatic techniques. For example, a “live story” may constitute a curated stream of user-submitted content from various locations and events. Users whose client devices have location services enabled and are at a common location event at a particular time may, for example, be presented with an option, via a user interface of the interaction client **104**, to contribute content to a particular live story. The live story may be identified to the user by the interaction client **104**, based on his or her location. The end result is a “live story” told from a community perspective.

[0073] A further type of content collection is known as a “location story,” which enables a user whose user system **102** is located within a specific geographic location (e.g., on a college or university campus) to contribute to a particular collection. In some examples, a contribution to a location story may employ a second degree of authentication to verify that the end-user belongs to a specific organization or other entity (e.g., is a student on the university campus).

[0074] As mentioned above, the video table **314** stores video data that, in some examples, is associated with messages for which records are maintained within the message table **306**. Similarly, the image table **316** stores image data associated with messages for which message data is stored in the entity table **308**. The entity table **308** may associate various augmentations from the augmentation table **312** with various images and videos stored in the image table **316** and the video table **314**.

[0075] The databases **304** also include trained machine learning (ML) technique(s) **307** that stores parameters of one

or more machine learning models that have been trained during training of the fashion item transfer system **500**. For example, trained machine learning techniques **307** stores the trained parameters of one or more artificial neural network machine learning models or techniques.

Data Communications Architecture

[0076] FIG. 4 is a schematic diagram illustrating a structure of a message **400**, according to some examples, generated by an interaction client **104** for communication to a further interaction client **104** via the interaction servers **124**. The content of a particular message **400** is used to populate the message table **306** stored within the database **304**, accessible by the interaction servers **124**. Similarly, the content of a message **400** is stored in memory as “in-transit” or “in-flight” data of the user system **102** or the interaction servers **124**. A message **400** is shown to include the following example components:

[0077] Message identifier **402**: a unique identifier that identifies the message **400**.

[0078] Message text payload **404**: text, to be generated by a user via a user interface of the user system **102**, and that is included in the message **400**.

[0079] Message image payload **406**: image data, captured by a camera component of a user system **102** or retrieved from a memory component of a user system **102**, and that is included in the message **400**. Image data for a sent or received message **400** may be stored in the image table **316**.

[0080] Message video payload **408**: video data, captured by a camera component or retrieved from a memory component of the user system **102**, and that is included in the message **400**. Video data for a sent or received message **400** may be stored in the image table **316**.

[0081] Message audio payload **410**: audio data, captured by a microphone or retrieved from a memory component of the user system **102**, and that is included in the message **400**.

[0082] Message augmentation data **412**: augmentation data (e.g., filters, stickers, or other annotations or enhancements) that represents augmentations to be applied to message image payload **406**, message video payload **408**, or message audio payload **410** of the message **400**. Augmentation data for a sent or received message **400** may be stored in the augmentation table **312**.

[0083] Message duration parameter **414**: parameter value indicating, in seconds, the amount of time for which content of the message (e.g., the message image payload **406**, message video payload **408**, message audio payload **410**) is to be presented or made accessible to a user via the interaction client **104**.

[0084] Message geolocation parameter **416**: geolocation data (e.g., latitudinal and longitudinal coordinates) associated with the content payload of the message. Multiple message geolocation parameter **416** values may be included in the payload, each of these parameter values being associated with respect to content items included in the content (e.g., a specific image within the message image payload **406**, or a specific video in the message video payload **408**).

[0085] Message story identifier **418**: identifier values identifying one or more content collections (e.g., “sto-

ries” identified in the story table 318) with which a particular content item in the message image payload 406 of the message 400 is associated. For example, multiple images within the message image payload 406 may each be associated with multiple content collections using identifier values.

[0086] Message tag 420: each message 400 may be tagged with multiple tags, each of which is indicative of the subject matter of content included in the message payload. For example, where a particular image included in the message image payload 406 depicts an animal (e.g., a lion), a tag value may be included within the message tag 420 that is indicative of the relevant animal. Tag values may be generated manually, based on user input, or may be automatically generated using, for example, image recognition.

[0087] Message sender identifier 422: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system 102 on which the message 400 was generated and from which the message 400 was sent.

[0088] Message receiver identifier 424: an identifier (e.g., a messaging system identifier, email address, or device identifier) indicative of a user of the user system 102 to which the message 400 is addressed.

[0089] The contents (e.g., values) of the various components of message 400 may be pointers to locations in tables within which content data values are stored. For example, an image value in the message image payload 406 may be a pointer to (or address of) a location within an image table 316. Similarly, values within the message video payload 408 may point to data stored within an image table 316, values stored within the message augmentation data 412 may point to data stored in an augmentation table 312, values stored within the message story identifier 418 may point to data stored in a story table 318, and values stored within the message sender identifier 422 and the message receiver identifier 424 may point to user records stored within an entity table 308.

Friend Try-On AR System

[0090] FIG. 5 is a block diagram showing an example friend try-on AR system 500, according to example examples. The friend try-on AR system 500 includes a set of components 510 that operate on a set of input data (e.g., an image 501 including one or more frames depicting a person in a real-world environment). The friend try-on AR system 500 includes an image selection module 512, a person detection module 514, an avatar generation module 516, a fashion item selection module 517, an image modification module 518, and an image display module 520. All or some of the components of the friend try-on AR system 500 can be implemented by a server, in which case, the image 501 are provided to the server by the user system 102. In some cases, some or all of the components of the friend try-on AR system 500 can be implemented by the user system 102 or can be distributed across a set of user systems 102.

[0091] In some examples, the friend try-on AR system 500 accesses a plurality of images that depict one or more persons. The friend try-on AR system 500 receives input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images and extracts features of the given person depicted in the individual image. The friend try-on AR system 500 applies

a machine learning model to the extracted features of the given person to generate an avatar that resembles the given person. The friend try-on AR system 500 applies one or more AR fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items.

[0092] In some examples, the friend try-on AR system 500 presents a list of AR fashion items. The friend try-on AR system 500 selects the one or more AR fashion items from the list of AR fashion items and applies the one or more AR fashion items to the avatar in response to selecting the one or more AR fashion items from the list of AR fashion items. In some examples, the avatar includes a three-dimensional (3D) avatar. In such cases, the friend try-on AR system 500 receives input that moves the avatar to which the one or more AR fashion items have been applied in 3D.

[0093] In some examples, the friend try-on AR system 500 presents the individual image of the plurality of images on the device to a user. The friend try-on AR system 500 detects a touch input of a portion of the individual image that depicts the given person to identify the given person. In some examples, the individual image depicts both the given person and the user.

[0094] In some examples, the friend try-on AR system 500 searches the plurality of images for additional images that depict different views of the given person. The friend try-on AR system 500 applies the machine learning model to the individual image and the additional images to generate the avatar of the given person. In some examples, the friend try-on AR system 500 predicts, by the machine learning model, a body shape and size of the given person based on the individual image and the additional images.

[0095] In some examples, the image provides an approximation of how one or more real-world fashion items corresponding to the one or more AR fashion items fit on the given person. In some examples, the image provides an approximation of how one or more real-world fashion items corresponding to the one or more AR fashion items fit on a particular person having the body shape and size of the given person depicted in the individual image.

[0096] In some examples, the friend try-on AR system 500 receives additional input that adjusts one or more parameters of the avatar. The friend try-on AR system 500 updates the avatar based on the adjusted one or more parameters. In some examples, the one or more parameters include at least one of a height, skin tone, hair color, body type, or body size.

[0097] In some examples, the machine learning model includes a generative artificial neural network (GAN). In some examples, the friend try-on AR system 500 trains the GAN by performing training operations including receiving training data including plurality of training images that depict a plurality of training persons and corresponding ground-truth avatars that resemble the training persons and applying the GAN to a first training image of the plurality of training images to generate an estimated avatar corresponding to a given training person depicted in the first training image. The training operations include retrieving the ground-truth avatar that resembles the given training person and computing a deviation between the ground-truth avatar that resembles the given training person and the estimated avatar corresponding to the given training person depicted in the first training image. The training operations include updating one or more parameters of the GAN based on the computed deviation.

[0098] In some examples, the friend try-on AR system 500 trains the GAN by performing training operations including receiving training data including plurality of training images that depict a plurality of training persons and corresponding ground-truth body shapes and sizes corresponding to the training persons and applying the GAN to a first training image of the plurality of training images to generate an estimated body shape and size corresponding to a given training person depicted in the first training image. The training operations include retrieving the ground-truth body shape and size of the given training person and computing a deviation between the ground-truth body shape and size and the estimated body shape and size. The training operations include updating one or more parameters of the GAN based on the computed deviation.

[0099] In some examples, the friend try-on AR system 500 accesses a communication session including a plurality of messages exchanged between a first client device of a user and a second client device of the given person. The plurality of messages can include one or more images that depict the user and the given person. The friend try-on AR system 500 selects the plurality of images based on the plurality of messages.

[0100] In some examples, the friend try-on AR system 500 shares the image that resembles the given person wearing the one or more AR fashion items to a specified recipient.

[0101] For example, the image selection module 512 can receive a user request to launch a friend try-on shopping experience. In response, the image selection module 512 accesses a list of conversations that the user of the user system 102 has previously been engaged in. The image selection module 512 can receive input that selects a particular conversation associated with a target person with whom the user has previously exchanged one or more messages. The messages may have included one or more images that depict the target person alone or with the user and/or one or more other people.

[0102] The image selection module 512 can present the one or more images to the user. The image selection module 512 can receive input from the user that identifies or selects the target person who is depicted in the one or more images. For example, the image selection module 512 can receive a touch input that taps or touches a portion of one or more of the images in a region in which the target person is depicted. In response, the image selection module 512 extracts the portion of the image that depicts the target person. The image selection module 512 generates or estimates one or more features of the target person, such as a look of the person including skin tone, body type, body style, weight, height, hair color, and so forth.

[0103] Specifically, as shown in FIG. 6, an image 600 is presented on the user system 102. The image 600 depicts a plurality of persons 610 and 620. The image selection module 512 receives input that taps or otherwise identifies a target person 610 from the plurality of persons. The image selection module 512 extracts or crops out a region of the image 600 that includes a depiction of the target person 610. The image selection module 512 provides that cropped out region to the person detection module 514 and/or generates one or more features of the target person 610.

[0104] In some examples, the image selection module 512 uses the one or more features to search other images that are stored on the user system 102. The image selection module 512 can select a portion of the other images that include

depictions of persons having similar or the same one or more features as the target person. The image selection module 512 assembles a collection of images that include the one or more images that have been used to identify the target person and the portion of the plurality of other images that include depictions of the persons having the similar or the same one or more features as the target person. In some cases, the image selection module 512 communicates the one or more images (or the extracted portion of the one or more images) to the person detection module 514 to detect and generate the one or more features.

[0105] The person detection module 514 implements one or more machine learning models (e.g., one or more neural networks) that have been trained to detect and track one or more persons and/or body parts in one or more images. For example, during training, the machine learning model of the person detection module 514 receives a given training image (or video) from a plurality of training images (or videos) that depict one or more persons and/or body parts. The plurality of training images is associated with corresponding ground truth markers or indications of the features (e.g., weight, height, hair color, skin tones, and so forth) of the persons and/or body parts depicted in the training images and can be received or accessed from training image data stored in data structures 300. The person detection module 514 applies one or more machine learning models to a given training image. The person detection module 514 generates estimated features for the persons and/or body parts depicted in the given training image.

[0106] The person detection module 514 obtains a known or predetermined ground-truth feature corresponding to the given training image. The person detection module 514 compares (computes a deviation between) the estimated features with the ground truth features. Based on a difference threshold of the comparison (or deviation), the person detection module 514 updates one or more coefficients or parameters and obtains one or more additional training images from the training data.

[0107] After a specified number of epochs or batches of training images have been processed and/or when a difference threshold (or deviation) (computed as a function of a difference or deviation between the estimated features and the ground-truth features) reaches a specified value, the person detection module 514 completes training, and the parameters and coefficients of the person detection module 514 are stored as a trained machine learning technique.

[0108] In some examples, the person detection module 514 presents the features of the target person to a user in a list. The person detection module 514 can receive input from a user that modifies and/or adds features or parameters to the list of features. For example, the person detection module 514 can receive input that changes a body type and/or hair color to match the target person. The person detection module 514 provides the features of the target person to the avatar generation module 516.

[0109] The avatar generation module 516 includes a generative machine learning model, such as a generative artificial neural network (GAN). The GAN processes the features of the target person to generate an avatar or avatar model representing and/or resembling features of the target person. In some cases, the GAN is trained to generate models based on training data. Specifically, during training, the GAN receives a given training image (or video) and/or features of a training person from a plurality of training

images (or videos) and/or features of a training person. The plurality of training images is associated with corresponding ground truth avatars of the training persons. The GAN is applied to a given training image and/or features of the training person. The GAN generates an avatar or avatar model (e.g., 3D avatar model) that resembles the training person.

[0110] The GAN obtains a known or predetermined ground-truth avatar or avatar model, such as a 3D avatar model. The GAN compares (computes a deviation between) the estimated avatar or avatar model with the ground truth avatar or avatar model. Based on a difference threshold of the comparison (or deviation), the GAN updates one or more coefficients or parameters and obtains one or more additional training images from the training data.

[0111] After a specified number of epochs or batches of training images have been processed and/or when a difference threshold (or deviation) (computed as a function of a difference or deviation between the estimated avatar and the ground-truth avatar) reaches a specified value, the GAN completes training, and the parameters and coefficients of the GAN are stored as a trained machine learning technique.

[0112] In some examples, the avatar generation module 516 presents the avatar or 3D avatar to the user on the user system 102 before and/or after overlaying the avatar with one or more AR fashion items. The avatar generation module 516 can receive input that modifies parameters of the avatar (e.g., a height, weight, body type, skin tone, and so forth). The avatar generation module 516 adjusts the look of the avatar based on the received input to update how the avatar looks wearing one or more AR fashion items.

[0113] For example, as shown in FIG. 8, the avatar generation module 516 presents a GUI 800. The GUI 800 includes an avatar region 810 that includes the 3D avatar of the target person (e.g., friend) that was generated based on the one or more images, such as image 600. The second GUI 800 also includes a parameters region 820. The parameters region 820 lists various avatar parameters of the avatar depicted in the avatar region 810. The parameters region 820 can receive input from a user that adjusts, deletes, or adds parameters. Namely, the input can modify a body style and/or hair color of the avatar depicted in the avatar region 810. In response, the avatar generation module 516 updates the look of the avatar depicted in the avatar region 810. In some cases, the avatar generation module 516 updates the avatar that is presented in other GUIs, such as the GUI 700, shown in FIG. 7.

[0114] In some cases, the fashion item selection module 517 presents a list of available AR fashion items to a user of the user system 102. In some examples, the fashion item selection module 517 receives a selection of a given AR fashion item or multiple AR fashion items from the list of AR fashion items. The fashion item selection module 517 can access or retrieve the AR fashion item(s). The fashion item selection module 517 provides the retrieved AR fashion item(s) to the avatar generation module 516. In some cases, the fashion item selection module 517 presents an online store GUI to a user which features many different fashion items. The fashion item selection module 517 receives a user selection of a particular fashion item from the GUI and searches the internet to find an AR version of the selected fashion item. The fashion item selection module 517 retrieves the AR version of the fashion item and provides the AR version to the avatar generation module 516.

[0115] The avatar generation module 516 receives the AR version of the first fashion item and overlays the AR version of the first fashion item on top of the avatar of the target person generated based on the one or more images stored on the user system 102. In some cases, the avatar generation module 516 uses tracking information for the body parts or person depicted in the avatar to continuously adjust a position of the AR version of the fashion item on the avatar. This results in a realistic display that depicts the avatar of the target person in the real-time video feed wearing the AR version of the fashion item. Namely, the display provides an approximation of how one or more real-world fashion items corresponding to the one or more AR fashion items fit on the target person (e.g., friend of the user).

[0116] In some cases, the display provides an approximation of how one or more real-world fashion items corresponding to the one or more AR fashion items fit on a particular person having the body shape and size of the target person depicted in the one or more images. For example, the user may not have in their possession images of an individual for whom they would like to shop. In such cases, the user can select the target person depicted in the one or more images that has a similar look and/or body type as the individual for whom they would like to shop. In these circumstances, the avatar that is generated for the target person can mimic how the fashion item will look on the individual who has the same or similar body type and features as the target person.

[0117] FIG. 7 shows an illustrative GUI 700 combined with an AR experience, according to some examples. Specifically, the friend try-on AR system 500 receives a user request to access the GUI 700, such as in response to receiving a user selection of a portal associated with a merchant (e.g., a website) that lists various fashion items and/or AR fashion items. The GUI 700 presents a 2D image that includes images of one or more fashion items 720. The friend try-on AR system 500 selects a first fashion item 722 from the one or more fashion items 720 depicted in the GUI 700, such as automatically or in response to a user input that taps or selects the first fashion item 722. The friend try-on AR system 500 generates or accesses a first AR fashion item corresponding to the first fashion item 722 (e.g., the first fashion item image).

[0118] The friend try-on AR system 500 receives a 3D avatar or avatar model from the avatar generation module 516 and presents the 3D avatar 730 in the avatar region 710. The friend try-on AR system 500 applies the first AR fashion item 732 (or a portion of the first AR fashion item) on the 3D avatar 730 or avatar model. The avatar region 710 can be a window, such as a circular window on square window, on the GUI 700. In this way, the avatar region 710 is presented together with the one or more fashion items 720 depicted in the GUI.

[0119] The friend try-on AR system 500 can receive input from the user that selects any of the fashion items depicted in the GUI 700. For example, the user can tap a different fashion item image depicted in the second GUI 700. In a similar manner, the friend try-on AR system 500 updates the 3D avatar 730 to present the AR version of the fashion item corresponding to the fashion item selected by the user instead of the AR version of the previously selected fashion item. The AR system 224 can receive input that rotates or moves the 3D avatar 730 in 3D to visualize how the AR

fashion item looks on the 3D avatar **730** from different views, angles and perspectives.

[0120] FIG. 9 is a flowchart of a process **900**, in accordance with some examples. Although the flowchart describes the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be re-arranged. A process is terminated when its operations are completed. A process may correspond to a method, a procedure, and the like. The steps of methods may be performed in whole or in part, may be performed in conjunction with some or all of the steps in other methods, and may be performed by any number of different systems or any portion thereof, such as a processor included in any of the systems.

[0121] At operation **901**, the friend try-on AR system **500** (e.g., a server or user system **102**) accesses a plurality of images that depict one or more persons, as discussed above.

[0122] At operation **902**, the friend try-on AR system **500** receives input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images, as discussed above.

[0123] At operation **903**, the friend try-on AR system **500** extracts features of the given person depicted in the individual image, as discussed above.

[0124] At operation **904**, the friend try-on AR system **500** applies a machine learning model to the extracted features of the given person to generate an avatar that resembles the given person, as discussed above.

[0125] At operation **905**, the friend try-on AR system **500** applies one or more augmented reality (AR) fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items, as discussed above.

Machine Architecture

[0126] FIG. 10 is a diagrammatic representation of the machine **1000** within which instructions **1002** (e.g., software, a program, an application, an applet, an app, or other executable code) for causing the machine **1000** to perform any one or more of the methodologies discussed herein may be executed. For example, the instructions **1002** may cause the machine **1000** to execute any one or more of the methods described herein. The instructions **1002** transform the general, non-programmed machine **1000** into a particular machine **1000** programmed to carry out the described and illustrated functions in the manner described. The machine **1000** may operate as a standalone device or may be coupled (e.g., networked) to other machines. In a networked deployment, the machine **1000** may operate in the capacity of a server machine or a client machine in a server-client network environment, or as a peer machine in a peer-to-peer (or distributed) network environment. The machine **1000** may comprise, but not be limited to, a server computer, a client computer, a personal computer (PC), a tablet computer, a laptop computer, a netbook, a set-top box (STB), a personal digital assistant (PDA), an entertainment media system, a cellular telephone, a smartphone, a mobile device, a wearable device (e.g., a smartwatch), a smart home device (e.g., a smart appliance), other smart devices, a web appliance, a network router, a network switch, a network bridge, or any machine capable of executing the instructions **1002**, sequentially or otherwise, that specify actions to be taken by the machine **1000**. Further, while a single machine **1000** is illustrated, the term “machine” shall also be taken to include

a collection of machines that individually or jointly execute the instructions **1002** to perform any one or more of the methodologies discussed herein. The machine **1000**, for example, may comprise the user system **102** or any one of multiple server devices forming part of the interaction server system **110**. In some examples, the machine **1000** may also comprise both client and server systems, with certain operations of a particular method or algorithm being performed on the server-side and with certain operations of the particular method or algorithm being performed on the client-side.

[0127] The machine **1000** may include processors **1004**, memory **1006**, and input/output I/O components **1008**, which may be configured to communicate with each other via a bus **1010**. In an example, the processors **1004** (e.g., a Central Processing Unit (CPU), a Reduced Instruction Set Computing (RISC) Processor, a Complex Instruction Set Computing (CISC) Processor, a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Radio-Frequency Integrated Circuit (RFIC), another processor, or any suitable combination thereof) may include, for example, a processor **1012** and a processor **1014** that execute the instructions **1002**. The term “processor” is intended to include multi-core processors that may comprise two or more independent processors (sometimes referred to as “cores”) that may execute instructions contemporaneously. Although FIG. 10 shows multiple processors **1004**, the machine **1000** may include a single processor with a single-core, a single processor with multiple cores (e.g., a multi-core processor), multiple processors with a single core, multiple processors with multiples cores, or any combination thereof.

[0128] The memory **1006** includes a main memory **1016**, a static memory **1018**, and a storage unit **1020**, both accessible to the processors **1004** via the bus **1010**. The main memory **1006**, the static memory **1018**, and storage unit **1020** store the instructions **1002** embodying any one or more of the methodologies or functions described herein. The instructions **1002** may also reside, completely or partially, within the main memory **1016**, within the static memory **1018**, within machine-readable medium **1022** within the storage unit **1020**, within at least one of the processors **1004** (e.g., within the processor’s cache memory), or any suitable combination thereof, during execution thereof by the machine **1000**.

[0129] The I/O components **1008** may include a wide variety of components to receive input, provide output, produce output, transmit information, exchange information, capture measurements, and so on. The specific I/O components **1008** that are included in a particular machine will depend on the type of machine. For example, portable machines such as mobile phones may include a touch input device or other such input mechanisms, while a headless server machine will likely not include such a touch input device. It will be appreciated that the I/O components **1008** may include many other components that are not shown in FIG. 10. In various examples, the I/O components **1008** may include user output components **1024** and user input components **1026**. The user output components **1024** may include visual components (e.g., a display such as a plasma display panel (PDP), a light-emitting diode (LED) display, a liquid crystal display (LCD), a projector, or a cathode ray tube (CRT)), acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor, resistance mechanisms), other signal generators, and so forth. The user input

components **1026** may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or another pointing instrument), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0130] In further examples, the I/O components **1008** may include biometric components **1028**, motion components **1030**, environmental components **1032**, or position components **1034**, among a wide array of other components. For example, the biometric components **1028** include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram-based identification), and the like. The motion components **1030** include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope).

[0131] The environmental components **1032** include, for example, one or cameras (with still image/photograph and video capabilities), illumination sensor components (e.g., photometer), temperature sensor components (e.g., one or more thermometers that detect ambient temperature), humidity sensor components, pressure sensor components (e.g., barometer), acoustic sensor components (e.g., one or more microphones that detect background noise), proximity sensor components (e.g., infrared sensors that detect nearby objects), gas sensors (e.g., gas detection sensors to detection concentrations of hazardous gases for safety or to measure pollutants in the atmosphere), or other components that may provide indications, measurements, or signals corresponding to a surrounding physical environment.

[0132] With respect to cameras, the user system **102** may have a camera system comprising, for example, front cameras on a front surface of the user system **102** and rear cameras on a rear surface of the user system **102**. The front cameras may, for example, be used to capture still images and video of a user of the user system **102** (e.g., “selfies”), which may then be augmented with augmentation data (e.g., filters) described above. The rear cameras may, for example, be used to capture still images and videos in a more traditional camera mode, with these images similarly being augmented with augmentation data. In addition to front and rear cameras, the user system **102** may also include a 360° camera for capturing 360° photographs and videos.

[0133] Further, the camera system of the user system **102** may include dual rear cameras (e.g., a primary camera as well as a depth-sensing camera), or even triple, quad or penta rear camera configurations on the front and rear sides of the user system **102**. These multiple cameras systems may include a wide camera, an ultra-wide camera, a telephoto camera, a macro camera, and a depth sensor, for example.

[0134] The position components **1034** include location sensor components (e.g., a GPS receiver component), altitude sensor components (e.g., altimeters or barometers that

detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like.

[0135] Communication may be implemented using a wide variety of technologies. The I/O components **1008** further include communication components **1036** operable to couple the machine **1000** to a network **1038** or devices **1040** via respective coupling or connections. For example, the communication components **1036** may include a network interface component or another suitable device to interface with the network **1038**. In further examples, the communication components **1036** may include wired communication components, wireless communication components, cellular communication components, Near Field Communication (NFC) components, Bluetooth® components (e.g., Bluetooth® Low Energy), Wi-Fi® components, and other communication components to provide communication via other modalities. The devices **1040** may be another machine or any of a wide variety of peripheral devices (e.g., a peripheral device coupled via a USB).

[0136] Moreover, the communication components **1036** may detect identifiers or include components operable to detect identifiers. For example, the communication components **1036** may include Radio Frequency Identification (RFID) tag reader components, NFC smart tag detection components, optical reader components (e.g., an optical sensor to detect one-dimensional bar codes such as Universal Product Code (UPC) bar code, multi-dimensional bar codes such as Quick Response (QR) code, Aztec code, Data Matrix, Dataglyph™, MaxiCode, PDF417, Ultra Code, UCC RSS-2D bar code, and other optical codes), or acoustic detection components (e.g., microphones to identify tagged audio signals). In addition, a variety of information may be derived via the communication components **1036**, such as location via Internet Protocol (IP) geolocation, location via Wi-Fi® signal triangulation, location via detecting an NFC beacon signal that may indicate a particular location, and so forth.

[0137] The various memories (e.g., main memory **1016**, static memory **1018**, and memory of the processors **1004**) and storage unit **1020** may store one or more sets of instructions and data structures (e.g., software) embodying or used by any one or more of the methodologies or functions described herein. These instructions (e.g., the instructions **1002**), when executed by processors **1004**, cause various operations to implement the disclosed examples.

[0138] The instructions **1002** may be transmitted or received over the network **1038**, using a transmission medium, via a network interface device (e.g., a network interface component included in the communication components **1036**) and using any one of several well-known transfer protocols (e.g., hypertext transfer protocol (HTTP)). Similarly, the instructions **1002** may be transmitted or received using a transmission medium via a coupling (e.g., a peer-to-peer coupling) to the devices **1040**.

Software Architecture

[0139] FIG. 11 is a block diagram **1100** illustrating a software architecture **1102**, which can be installed on any one or more of the devices described herein. The software architecture **1102** is supported by hardware such as a machine **1104** that includes processors **1106**, memory **1108**, and I/O components **1110**. In this example, the software

architecture **1102** can be conceptualized as a stack of layers, where each layer provides a particular functionality. The software architecture **1102** includes layers such as an operating system **1112**, libraries **1114**, frameworks **1116**, and applications **1118**. Operationally, the applications **1118** invoke API calls **1120** through the software stack and receive messages **1122** in response to the API calls **1120**.

[0140] The operating system **1112** manages hardware resources and provides common services. The operating system **1112** includes, for example, a kernel **1124**, services **1126**, and drivers **1128**. The kernel **1124** acts as an abstraction layer between the hardware and the other software layers. For example, the kernel **1124** provides memory management, processor management (e.g., scheduling), component management, networking, and security settings, among other functionalities. The services **1126** can provide other common services for the other software layers. The drivers **1128** are responsible for controlling or interfacing with the underlying hardware. For instance, the drivers **1128** can include display drivers, camera drivers, BLUETOOTH® or BLUETOOTH® Low Energy drivers, flash memory drivers, serial communication drivers (e.g., USB drivers), WI-FI® drivers, audio drivers, power management drivers, and so forth.

[0141] The libraries **1114** provide a common low-level infrastructure used by the applications **1118**. The libraries **1114** can include system libraries **1130** (e.g., C standard library) that provide functions such as memory allocation functions, string manipulation functions, mathematic functions, and the like. In addition, the libraries **1114** can include API libraries **1132** such as media libraries (e.g., libraries to support presentation and manipulation of various media formats such as Moving Picture Experts Group-4 (MPEG4), Advanced Video Coding (H.264 or AVC), Moving Picture Experts Group Layer-3 (MP3), Advanced Audio Coding (AAC), Adaptive Multi-Rate (AMR) audio codec, Joint Photographic Experts Group (JPEG or JPG), or Portable Network Graphics (PNG)), graphics libraries (e.g., an OpenGL framework used to render in two dimensions (2D) and three dimensions (3D) in a graphic content on a display), database libraries (e.g., SQLite to provide various relational database functions), web libraries (e.g., WebKit to provide web browsing functionality), and the like. The libraries **1114** can also include a wide variety of other libraries **1134** to provide many other APIs to the applications **1118**.

[0142] The frameworks **1116** provide a common high-level infrastructure that is used by the applications **1118**. For example, the frameworks **1116** provide various graphical user interface (GUI) functions, high-level resource management, and high-level location services. The frameworks **1116** can provide a broad spectrum of other APIs that can be used by the applications **1118**, some of which may be specific to a particular operating system or platform.

[0143] In an example, the applications **1118** may include a home application **1136**, a contacts application **1138**, a browser application **1140**, a book reader application **1142**, a location application **1144**, a media application **1146**, a messaging application **1148**, a game application **1150**, and a broad assortment of other applications such as a third-party application **1152**. The applications **1118** are programs that execute functions defined in the programs. Various programming languages can be employed to create one or more of the applications **1118**, structured in a variety of manners, such as object-oriented programming languages (e.g.,

Objective-C, Java, or C++) or procedural programming languages (e.g., C or assembly language). In a specific example, the third-party application **1152** (e.g., an application developed using the ANDROID™ or IOS™ software development kit (SDK) by an entity other than the vendor of the particular platform) may be mobile software running on a mobile operating system such as IOS™, ANDROID™, WINDOWS® Phone, or another mobile operating system. In this example, the third-party application **1152** can invoke the API calls **1120** provided by the operating system **1112** to facilitate functionalities described herein.

System With Head-Wearable Apparatus

[0144] FIG. 12 illustrates a system **1200** including a head-wearable apparatus **116** with a selector input device, according to some examples. FIG. 12 is a high-level functional block diagram of an example head-wearable apparatus **116** communicatively coupled to a mobile device **114** and various server systems **1204** (e.g., the interaction server system **110**) via various networks **108**.

[0145] The head-wearable apparatus **116** includes one or more cameras, each of which may be, for example, a visible light camera **1206**, an infrared emitter **1208**, and an infrared camera **1210**.

[0146] The mobile device **114** connects with head-wearable apparatus **116** using both a low-power wireless connection **1212** and a high-speed wireless connection **1214**. The mobile device **114** is also connected to the server system **1204** and the network **1216**.

[0147] The head-wearable apparatus **116** further includes two image displays of the image display of optical assembly **1218**. The two image displays of optical assembly **1218** include one associated with the left lateral side and one associated with the right lateral side of the head-wearable apparatus **116**. The head-wearable apparatus **116** also includes an image display driver **1220**, an image processor **1222**, low-power circuitry **1224**, and high-speed circuitry **1226**. The image display of optical assembly **1218** is for presenting images and videos, including an image that can include a graphical user interface to a user of the head-wearable apparatus **116**.

[0148] The image display driver **1220** commands and controls the image display of optical assembly **1218**. The image display driver **1220** may deliver image data directly to the image display of optical assembly **1218** for presentation or may convert the image data into a signal or data format suitable for delivery to the image display device. For example, the image data may be video data formatted according to compression formats, such as H.264 (MPEG-4 Part 10), HEVC, Theora, Dirac, RealVideo RV40, VP8, VP9, or the like, and still image data may be formatted according to compression formats such as Portable Network Group (PNG), Joint Photographic Experts Group (JPEG), Tagged Image File Format (TIFF) or exchangeable image file format (EXIF) or the like.

[0149] The head-wearable apparatus **116** includes a frame and stems (or temples) extending from a lateral side of the frame. The head-wearable apparatus **116** further includes a user input device **1228** (e.g., touch sensor or push button), including an input surface on the head-wearable apparatus **116**. The user input device **1228** (e.g., touch sensor or push button) is to receive from the user an input selection to manipulate the graphical user interface of the presented image.

[0150] The components shown in FIG. 12 for the head-wearable apparatus 116 are located on one or more circuit boards, for example a PCB or flexible PCB, in the rims or temples. Alternatively, or additionally, the depicted components can be located in the chunks, frames, hinges, or bridge of the head-wearable apparatus 116. Left and right visible light cameras 1206 can include digital camera elements such as a complementary metal oxide-semiconductor (CMOS) image sensor, charge-coupled device, camera lenses, or any other respective visible or light-capturing elements that may be used to capture data, including images of scenes with unknown objects.

[0151] The head-wearable apparatus 116 includes a memory 1202, which stores instructions to perform a subset or all of the functions described herein. The memory 1202 can also include storage device.

[0152] As shown in FIG. 12, the high-speed circuitry 1226 includes a high-speed processor 1230, a memory 1202, and high-speed wireless circuitry 1232. In some examples, the image display driver 1220 is coupled to the high-speed circuitry 1226 and operated by the high-speed processor 1230 in order to drive the left and right image displays of the image display of optical assembly 1218. The high-speed processor 1230 may be any processor capable of managing high-speed communications and operation of any general computing system needed for the head-wearable apparatus 116. The high-speed processor 1230 includes processing resources needed for managing high-speed data transfers on a high-speed wireless connection 1214 to a wireless local area network (WLAN) using the high-speed wireless circuitry 1232. In certain examples, the high-speed processor 1230 executes an operating system such as a LINUX operating system or other such operating system of the head-wearable apparatus 116, and the operating system is stored in the memory 1202 for execution. In addition to any other responsibilities, the high-speed processor 1230 executing a software architecture for the head-wearable apparatus 116 is used to manage data transfers with high-speed wireless circuitry 1232. In certain examples, the high-speed wireless circuitry 1232 is configured to implement Institute of Electrical and Electronic Engineers (IEEE) 802.11 communication standards, also referred to herein as WiFi. In some examples, other high-speed communications standards may be implemented by the high-speed wireless circuitry 1232.

[0153] The low-power wireless circuitry 1234 and the high-speed wireless circuitry 1232 of the head-wearable apparatus 116 can include short-range transceivers (Bluetooth™) and wireless wide, local, or wide area network transceivers (e.g., cellular or WiFi). Mobile device 114, including the transceivers communicating via the low-power wireless connection 1212 and the high-speed wireless connection 1214, may be implemented using details of the architecture of the head-wearable apparatus 116, as can other elements of the network 1216.

[0154] The memory 1202 includes any storage device capable of storing various data and applications, including, among other things, camera data generated by the left and right visible light cameras 1206, the infrared camera 1210, and the image processor 1222, as well as images generated for display by the image display driver 1220 on the image displays of the image display of optical assembly 1218. While the memory 1202 is shown as integrated with high-speed circuitry 1226, in some examples, the memory 1202 may be an independent standalone element of the head-

wearable apparatus 116. In certain such examples, electrical routing lines may provide a connection through a chip that includes the high-speed processor 1230 from the image processor 1222 or the low-power processor 1236 to the memory 1202. In some examples, the high-speed processor 1230 may manage addressing of the memory 1202 such that the low-power processor 1236 will boot the high-speed processor 1230 any time that a read or write operation involving memory 1202 is needed.

[0155] As shown in FIG. 12, the low-power processor 1236 or high-speed processor 1230 of the head-wearable apparatus 116 can be coupled to the camera (visible light camera 1206, infrared emitter 1208, or infrared camera 1210), the image display driver 1220, the user input device 1228 (e.g., touch sensor or push button), and the memory 1202.

[0156] The head-wearable apparatus 116 is connected to a host computer. For example, the head-wearable apparatus 116 is paired with the mobile device 114 via the high-speed wireless connection 1214 or connected to the server system 1204 via the network 1216. The server system 1204 may be one or more computing devices as part of a service or network computing system, for example, that includes a processor, a memory, and network communication interface to communicate over the network 1216 with the mobile device 114 and the head-wearable apparatus 116.

[0157] The mobile device 114 includes a processor and a network communication interface coupled to the processor. The network communication interface allows for communication over the network 1216, low-power wireless connection 1212, or high-speed wireless connection 1214. Mobile device 114 can further store at least portions of the instructions for generating binaural audio content in the mobile device 114's memory to implement the functionality described herein.

[0158] Output components of the head-wearable apparatus 116 include visual components, such as a display such as a liquid crystal display (LCD), a plasma display panel (PDP), a light-emitting diode (LED) display, a projector, or a waveguide. The image displays of the optical assembly are driven by the image display driver 1220. The output components of the head-wearable apparatus 116 further include acoustic components (e.g., speakers), haptic components (e.g., a vibratory motor), other signal generators, and so forth. The input components of the head-wearable apparatus 116, the mobile device 114, and server system 1204, such as the user input device 1228, may include alphanumeric input components (e.g., a keyboard, a touch screen configured to receive alphanumeric input, a photo-optical keyboard, or other alphanumeric input components), point-based input components (e.g., a mouse, a touchpad, a trackball, a joystick, a motion sensor, or other pointing instruments), tactile input components (e.g., a physical button, a touch screen that provides location and force of touches or touch gestures, or other tactile input components), audio input components (e.g., a microphone), and the like.

[0159] The head-wearable apparatus 116 may also include additional peripheral device elements. Such peripheral device elements may include biometric sensors, additional sensors, or display elements integrated with the head-wearable apparatus 116. For example, peripheral device elements may include any I/O components including output components, motion components, position components, or any other such elements described herein.

[0160] For example, the biometric components include components to detect expressions (e.g., hand expressions, facial expressions, vocal expressions, body gestures, or eye-tracking), measure biosignals (e.g., blood pressure, heart rate, body temperature, perspiration, or brain waves), identify a person (e.g., voice identification, retinal identification, facial identification, fingerprint identification, or electroencephalogram based identification), and the like. The motion components include acceleration sensor components (e.g., accelerometer), gravitation sensor components, rotation sensor components (e.g., gyroscope), and so forth. The position components include location sensor components to generate location coordinates (e.g., a Global Positioning System (GPS) receiver component), Wi-Fi or Bluetooth™ transceivers to generate positioning system coordinates, altitude sensor components (e.g., altimeters or barometers that detect air pressure from which altitude may be derived), orientation sensor components (e.g., magnetometers), and the like. Such positioning system coordinates can also be received over low-power wireless connections **1212** and high-speed wireless connection **1214** from the mobile device **114** via the low-power wireless circuitry **1234** or high-speed wireless circuitry **1232**.

GLOSSARY

[0161] “Carrier signal” refers, for example, to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device.

[0162] “Client device” refers, for example, to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

[0163] “Communication network” refers, for example, to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project

(3GPP) including 3G, fourth-generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

[0164] “Component” refers, for example, to a device, physical entity, or logic having boundaries defined by function or subroutine calls, branch points, APIs, or other technologies that provide for the partitioning or modularization of particular processing or control functions. Components may be combined via their interfaces with other components to carry out a machine process. A component may be a packaged functional hardware unit designed for use with other components and a part of a program that usually performs a particular function of related functions. Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components.

[0165] A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

[0166] A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an application-specific integrated circuit (ASIC). A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processors. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily configured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

[0167] Considering examples in which hardware components are temporarily configured (e.g., programmed), each of the hardware components need not be configured or instantiated at any one instance in time. For example, where a hardware component comprises a general-purpose processor configured by software to become a special-purpose processor, the general-purpose processor may be configured

as respectively different special-purpose processors (e.g., comprising different hardware components) at different times. Software accordingly configures a particular processor or processors, for example, to constitute a particular hardware component at one instance of time and to constitute a different hardware component at a different instance of time. Hardware components can provide information to, and receive information from, other hardware components. Accordingly, the described hardware components may be regarded as being communicatively coupled. Where multiple hardware components exist contemporaneously, communications may be achieved through signal transmission (e.g., over appropriate circuits and buses) between or among two or more of the hardware components. In examples in which multiple hardware components are configured or instantiated at different times, communications between such hardware components may be achieved, for example, through the storage and retrieval of information in memory structures to which the multiple hardware components have access. For example, one hardware component may perform an operation and store the output of that operation in a memory device to which it is communicatively coupled. A further hardware component may then, at a later time, access the memory device to retrieve and process the stored output. Hardware components may also initiate communications with input or output devices, and can operate on a resource (e.g., a collection of information). The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein.

[0168] As used herein, “processor-implemented component” refers to a hardware component implemented using one or more processors. Similarly, the methods described herein may be at least partially processor-implemented, with a particular processor or processors being an example of hardware. For example, at least some of the operations of a method may be performed by one or more processors or processor-implemented components. Moreover, the one or more processors may also operate to support performance of the relevant operations in a “cloud computing” environment or as a “software as a service” (SaaS). For example, at least some of the operations may be performed by a group of computers (as examples of machines including processors), with these operations being accessible via a network (e.g., the Internet) and via one or more appropriate interfaces (e.g., an API). The performance of certain of the operations may be distributed among the processors, not only residing within a single machine, but deployed across a number of machines. In some examples, the processors or processor-implemented components may be located in a single geographic location (e.g., within a home environment, an office environment, or a server farm). In other examples, the processors or processor-implemented components may be distributed across a number of geographic locations.

[0169] “Computer-readable storage medium” refers, for example, to both machine-storage media and transmission media. Thus, the terms include both storage devices/media and carrier waves/modulated data signals. The terms “machine-readable medium,” “computer-readable medium”

and “device-readable medium” mean the same thing and may be used interchangeably in this disclosure. “Ephemeral message” refers, for example, to a message that is accessible for a time-limited duration. An ephemeral message may be a text, an image, a video and the like. The access time for the ephemeral message may be set by the message sender. Alternatively, the access time may be a default setting or a setting specified by the recipient. Regardless of the setting technique, the message is transitory.

[0170] “Machine storage medium” refers, for example, to a single or multiple storage devices and media (e.g., a centralized or distributed database, and associated caches and servers) that store executable instructions, routines and data. The term shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, including memory internal or external to processors. Specific examples of machine-storage media, computer-storage media and device-storage media include non-volatile memory, including by way of example semiconductor memory devices, e.g., erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), FPGA, and flash memory devices; magnetic disks such as internal hard disks and removable disks; magneto-optical disks; and CD-ROM and DVD-ROM disks. The terms “machine-storage medium,” “device-storage medium,” “computer-storage medium” mean the same thing and may be used interchangeably in this disclosure.

[0171] The terms “machine-storage media,” “computer-storage media,” and “device-storage media” specifically exclude carrier waves, modulated data signals, and other such media, at least some of which are covered under the term “signal medium.” “Non-transitory computer-readable storage medium” refers, for example, to a tangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine. “Signal medium” refers, for example, to any intangible medium that is capable of storing, encoding, or carrying the instructions for execution by a machine and includes digital or analog communications signals or other intangible media to facilitate communication of software or data. The term “signal medium” shall be taken to include any form of a modulated data signal, carrier wave, and so forth. The term “modulated data signal” means a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. The terms “transmission medium” and “signal medium” mean the same thing and may be used interchangeably in this disclosure.

[0172] “User device” refers, for example, to a device accessed, controlled or owned by a user and with which the user interacts to perform an action, or an interaction with other users or computer systems. “Carrier signal” refers to any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the machine and includes digital or analog communications signals or other intangible media to facilitate communication of such instructions. Instructions may be transmitted or received over a network using a transmission medium via a network interface device. “Client device” refers to any machine that interfaces to a communications network to obtain resources from one or more server systems or other client devices. A client device may be, but is not limited to, a mobile phone, desktop computer, laptop, portable digital assistants (PDAs), smartphones, tablets, ultrabooks, netbooks, laptops, multi-

processor systems, microprocessor-based or programmable consumer electronics, game consoles, set-top boxes, or any other communication device that a user may use to access a network.

[0173] “Communication network” refers to one or more portions of a network that may be an ad hoc network, an intranet, an extranet, a virtual private network (VPN), a local area network (LAN), a wireless LAN (WLAN), a wide area network (WAN), a wireless WAN (WWAN), a metropolitan area network (MAN), the Internet, a portion of the Internet, a portion of the Public Switched Telephone Network (PSTN), a plain old telephone service (POTS) network, a cellular telephone network, a wireless network, a Wi-Fi® network, another type of network, or a combination of two or more such networks. For example, a network or a portion of a network may include a wireless or cellular network, and the coupling may be a Code Division Multiple Access (CDMA) connection, a Global System for Mobile communications (GSM) connection, or other types of cellular or wireless coupling. In this example, the coupling may implement any of a variety of types of data transfer technology, such as Single Carrier Radio Transmission Technology (1xRTT), Evolution-Data Optimized (EVDO) technology, General Packet Radio Service (GPRS) technology, Enhanced Data rates for GSM Evolution (EDGE) technology, third Generation Partnership Project (3GPP) including 3G, fourth-generation wireless (4G) networks, Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Worldwide Interoperability for Microwave Access (WiMAX), Long Term Evolution (LTE) standard, others defined by various standard-setting organizations, other long-range protocols, or other data transfer technology.

[0174] Components may constitute either software components (e.g., code embodied on a machine-readable medium) or hardware components. A “hardware component” is a tangible unit capable of performing certain operations and may be configured or arranged in a certain physical manner. In various examples, one or more computer systems (e.g., a standalone computer system, a client computer system, or a server computer system) or one or more hardware components of a computer system (e.g., a processor or a group of processors) may be configured by software (e.g., an application or application portion) as a hardware component that operates to perform certain operations as described herein.

[0175] A hardware component may also be implemented mechanically, electronically, or any suitable combination thereof. For example, a hardware component may include dedicated circuitry or logic that is permanently configured to perform certain operations. A hardware component may be a special-purpose processor, such as a field-programmable gate array (FPGA) or an ASIC. A hardware component may also include programmable logic or circuitry that is temporarily configured by software to perform certain operations. For example, a hardware component may include software executed by a general-purpose processor or other programmable processor. Once configured by such software, hardware components become specific machines (or specific components of a machine) uniquely tailored to perform the configured functions and are no longer general-purpose processors. It will be appreciated that the decision to implement a hardware component mechanically, in dedicated and permanently configured circuitry, or in temporarily config-

ured circuitry (e.g., configured by software), may be driven by cost and time considerations. Accordingly, the phrase “hardware component” (or “hardware-implemented component”) should be understood to encompass a tangible entity, be that an entity that is physically constructed, permanently configured (e.g., hardwired), or temporarily configured (e.g., programmed) to operate in a certain manner or to perform certain operations described herein.

[0176] The various operations of example methods described herein may be performed, at least partially, by one or more processors that are temporarily configured (e.g., by software) or permanently configured to perform the relevant operations. Whether temporarily or permanently configured, such processors may constitute processor-implemented components that operate to perform one or more operations or functions described herein.

[0177] Changes and modifications may be made to the disclosed examples without departing from the scope of the present disclosure. These and other changes or modifications are intended to be included within the scope of the present disclosure, as expressed in the following claims.

What is claimed is:

1. A method comprising:

applying a machine learning model comprising a generative artificial neural network (GAN) to extracted features of a given person to generate an avatar that resembles the given person, the machine learning model trained by performing training operations comprising:

receiving training data comprising a plurality of training images that depict a plurality of training persons and corresponding ground-truth avatars that resemble the plurality of training persons;

applying the machine learning model to a first training image of the plurality of training images to generate an estimated avatar corresponding to a given training person depicted in the first training image;

retrieving a ground-truth avatar that resembles the given training person;

computing a deviation between the ground-truth avatar that resembles the given training person and the estimated avatar corresponding to the given training person depicted in the first training image;

updating one or more parameters of the machine learning model based on the computed deviation; and

applying one or more augmented reality (AR) fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items.

2. The method of claim 1, further comprising:

accessing a plurality of images that depict one or more persons;

receiving input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images; and

extracting features of the given person depicted in the individual image.

3. The method of claim 1, wherein the avatar comprises a three-dimensional (3D) avatar, further comprising:

receiving input that moves the avatar to which the one or more AR fashion items have been applied in 3D.

4. The method of claim 1, further comprising:

presenting an individual image of a plurality of images on a device to a user; and

- detecting a touch input of a portion of the individual image that depicts the given person to identify the given person.
5. The method of claim 4, wherein the individual image depicts the given person and the user.
6. The method of claim 4, further comprising:
searching a plurality of images for additional images that depict different views of the given person; and
applying the machine learning model to the individual image and the additional images to generate the avatar of the given person.
7. The method of claim 6, further comprising:
predicting, by the machine learning model, a body shape and size of the given person based on the individual image and the additional images.
8. The method of claim 7, wherein the image provides an approximation of how one or more real-world fashion items corresponding to the one or more AR fashion items fit on the given person.
9. The method of claim 7, wherein the image provides an approximation of how one or more real-world fashion items corresponding to the one or more AR fashion items fit on a particular person having the body shape and size of the given person depicted in the individual image.
10. The method of claim 1, further comprising:
receiving additional input that adjusts one or more parameters of the avatar; and
updating the avatar based on the adjusted one or more parameters.
11. The method of claim 10, wherein the one or more parameters comprise at least one of a height, skin tone, hair color, body type, or body size.
12. The method of claim 1, further comprising training the GAN by performing training operations comprising:
receiving additional training data comprising ground-truth body shapes and sizes corresponding to the plurality of training persons;
applying the GAN to a second training image of the plurality of training images to generate an estimated body shape and size corresponding to a given training person depicted in the second training image;
retrieving a ground-truth body shape and size of the given training person;
computing a deviation between the ground-truth body shape and size and the estimated body shape and size; and
updating one or more parameters of the GAN based on the computed deviation.
13. The method of claim 1, further comprising:
accessing a communication session comprising a plurality of messages exchanged between a first client device of a user and a second client device of the given person, the plurality of messages comprising one or more images that depict the user and the given person; and
selecting the plurality of images based on the plurality of messages.
14. The method of claim 1, further comprising:
sharing the image that resembles the given person wearing the one or more AR fashion items to a specified recipient.
15. A system comprising:
at least one processor of a device configured to perform operations comprising:
applying a machine learning model comprising a generative artificial neural network (GAN) to extracted features of a given person to generate an avatar that resembles the given person, the machine learning model trained by performing training operations comprising:
receiving training data comprising a plurality of training images that depict a plurality of training persons and corresponding ground-truth avatars that resemble the plurality of training persons;
applying the machine learning model to a first training image of the plurality of training images to generate an estimated avatar corresponding to a given training person depicted in the first training image;
retrieving a ground-truth avatar that resembles the given training person;
computing a deviation between the ground-truth avatar that resembles the given training person and the estimated avatar corresponding to the given training person depicted in the first training image;
updating one or more parameters of the machine learning model based on the computed deviation; and
applying one or more augmented reality (AR) fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items.
16. The system of claim 15, the operations comprising:
accessing a plurality of images that depict one or more persons.
17. The system of claim 16, the operations comprising:
receiving input that identifies a given person of the one or more persons who is depicted in an individual image of the plurality of images; and
extracting features of the given person depicted in the individual image.
18. The system of claim 15, the operations comprising:
presenting a list of AR fashion items;
selecting the one or more AR fashion items from the list of AR fashion items; and
applying the one or more AR fashion items to the avatar in response to selecting the one or more AR fashion items from the list of AR fashion items.
19. The system of claim 15, wherein the avatar comprises a three-dimensional (3D) avatar, the operations comprising:
receiving input that moves the avatar to which the one or more AR fashion items have been applied in 3D.
20. A non-transitory machine-readable storage medium that includes instructions that, when executed by one or more processors of a device, cause the device to perform operations comprising:
applying a machine learning model comprising a generative artificial neural network (GAN) to extracted features of a given person to generate an avatar that resembles the given person, the machine learning model trained by performing training operations comprising:
receiving training data comprising a plurality of training images that depict a plurality of training persons and corresponding ground-truth avatars that resemble the plurality of training persons;

applying the machine learning model to a first training image of the plurality of training images to generate an estimated avatar corresponding to a given training person depicted in the first training image;
retrieving a ground-truth avatar that resembles the given training person;
computing a deviation between the ground-truth avatar that resembles the given training person and the estimated avatar corresponding to the given training person depicted in the first training image;
updating one or more parameters of the machine learning model based on the computed deviation; and
applying one or more augmented reality (AR) fashion items to the avatar to generate an image that resembles the given person wearing the one or more AR fashion items.

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